



Micro- and mesostructure-informed property spaces: Toward generalized approach to designing FGMs

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ABSTRACT

Understanding the feasible property space of micro- and mesostructures under varying parameters is essential for designing optimal engineered materials. This study delineates attainable stiffness-property spaces for microstructures using a novel constraint build-up method based on grain orientation distributions, and for mesostructures through geometry- and density-driven variations. The influence of feature-level degrees of freedom (DOF) on the resulting property-space topology is also revealed. Inconel 718 (IN-718) and polylactic acid (PLA) are selected as representative materials for microstructural and mesostructural grading, respectively, due to their structural versatility and manufacturing flexibility. Building on these results, a generalized functionally graded material (FGM) design framework is proposed that leverages micro- and mesostructural property constraints to enhance structural stiffness without requiring geometric modification. An iterative strain energy-based objective function is formulated to minimize structural deformation within the bounds of the defined property constraints. The method achieves smooth transitions of stiffness properties and their corresponding microstructure or mesostructure between adjacent elements, ensuring practical manufacturability of the FGM design. The optimized structure exhibits significant performance improvements over the baseline design. Furthermore, the framework's effectiveness is validated through numerical simulations and experimental testing of mesostructure graded specimens.

1. Introduction

Mitigating structural deformation under mechanical loading is a critical consideration across diverse engineering applications, from aerospace structures to biomedical implants [1–4]. This work introduces a scale-agnostic approach to exploring stiffness property spaces, where the same design strategy can be applied at different length scales. Within this viewpoint, we reveal an analogy in how property spaces arise at both the micro and meso scales, motivating a generalized approach to structural optimization for reducing deformation. We consider two distinct types of design variables: crystallographic texture at the microstructure level and unit-cell geometry at the mesostructure level. These two cases show that property spaces for stiffness parameters can be obtained in a similar manner across scales, even with different design variables. To illustrate this, we examine a metallic alloy microstructure to represent microscale texture design and a thermoplastic polymer architecture to represent mesoscale geometric design. The consistency of the resulting property closures across these scales supports the applicability of a unified optimization strategy.

Traditionally, approaches such as topology optimization have been widely used to improve structural performance by redistributing material and altering macroscopic geometry to achieve optimal load

paths [5,6]. However, topology optimization frequently leads to complex geometries that can be challenging or costly to manufacture [7], and the resultant designs may lack robustness under variable loading conditions [8]. In contrast, recent advancements in microstructure and mesostructure optimization provide a promising alternative by directly tailoring material properties at the microscopic or mesoscopic scales, including grain characteristics and variations in cellular unit cells, respectively, to effectively control anisotropic behavior [9–12]. Unlike topology optimization, microstructure and mesostructure optimization preserve the original structural geometry while enhancing performance through the control of intrinsic material characteristics to achieve targeted mechanical responses [13,14]. This approach allows for improved stiffness and efficiency without necessarily introducing complex, manufacturing-intensive geometries. Nevertheless, uncovering the behavior of micro- and meso-scale materials and the properties of their underlying elements requires an advanced understanding of physics-based materials modeling and computational design techniques.

In structural optimization research, two complementary approaches have been studied: macro-scale topology optimization focusing on

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Nomenclature

κ	Curvature vector
χ	Single crystal stiffness
γ	Shear strain
$\mathbf{A}$	Orientation distribution function
$\mathbf{B}$	Strain–displacement matrix
$\mathbf{C}$	Stiffness matrix
$\mathbf{F}$	Global force vector
$\mathbf{K}$	Global stiffness matrix
$\mathbf{n}$	Unit vector
$\mathbf{R}$	Lattice orientation matrix
$\mathbf{r}$	Rodrigues parameter
$\mathbf{U}$	Global displacement vector
ν	Poisson's ratio
ψ_i	Normalized strain energy density
ρ	Relative density
σ	Normal stress
τ	Shear stress
θ	Inclination angle of cell walls
ϵ	Normal strain
ϕ	Rotation angle in the Rodrigues orientation
E	Young's modulus
G	Shear modulus
J_n	Jacobian of n -th element
U_e	Strain energy
w_m	Integration weight of m -th integration point

geometry, and microstructure optimization targeting material architecture. Topology optimization is an established computational method for automatically generating high-performance structural layouts under given constraints [15], with foundations tracing back to the homogenization-based framework by Bendsøe and Kikuchi in 1988 [16], who modeled materials as periodic micro-cellular composites to compute effective properties. Subsequent studies in the 1990s shifted toward simple density-based formulations, such as Solid Isotropic Material with Penalization (SIMP), for purely geometric optimization yielding superior designs in aerospace and other fields [17–19]. Parallel efforts pursued intrinsic material optimization via tailored elements at micro- and meso-scales, showing that structural performance can be enhanced by designing periodic micro-architectures with extreme elastic properties like high stiffness or negative Poisson's ratio [20–23]. Functionally graded lattice structures, with gradually varying cell geometry or density, have been shown to significantly outperform uniform lattices in mechanical performance [24,25], as demonstrated by Wang et al. [26] achieving near-optimal multi-load designs through distorting periodic base-cell geometry. Both numerical simulations and experimental studies indicate that multiscale approaches outperform uniform-material structures in practice [27], with multi-physics extensions coupling mechanical objectives with thermal or fluid flow in lattice architectures [28,29].

In addition to geometry-based structural optimization, researchers have focused on microstructure optimization as a complementary strategy for enhancing mechanical performance by tailoring grain orientations, lattice architectures, and phase distributions at the microscale, achieving superior stiffness, strength, and anisotropic properties beyond homogeneous designs [6,30–34]. Pioneering work showed that optimized unit-cell topologies can realize extreme elastic moduli [35] and exotic behaviors like negative Poisson's ratios [36], illustrating the potential of architected materials. Modern frameworks enable concurrent optimization of macroscopic structure and material anisotropy,

integrating multi-scale design into structural optimization [37], with applications in advanced composites where aligning fiber orientations with load paths yields stiffness and strength gains [38], and in additively manufactured metals via site-specific control of crystallographic texture or lattice infill for lightweight structures [39,40]. Radman et al. [41] introduced the bi-directional evolutionary structural optimization (BESO) method to design spatially varying microstructures maximizing stiffness while controlling thermal conductivity in functionally graded materials (FGMs), enabling targeted multifunctional behavior with smooth gradation. Silva et al. [42] investigated stress-based topology optimization combined with multi-material additive manufacturing, demonstrating how FGMs can effectively reduce stress concentrations and enhance structural performance. In a subsequent study [43], they focused on topology optimization for thermoelastic structures, aiming to minimize elastic strain energy and von Mises stress using FGMs under thermomechanical loading. Both studies highlight the advantages of FGMs in optimizing structural performance under complex load conditions. Li et al. [44] developed a multiscale approach for graded cellular metamaterials, tailoring periodic microstructures for spatially graded stiffness and auxetic response. Conlan-Smith and James proposed [45] a framework optimizing structural layout and material gradation to reduce stress concentrations, lowering peak von Mises stress in components like L-brackets. Geng et al. [46] used laser additive manufacturing on Ti-6Al-4V to create gradients in grain structure and texture, resulting in varied elastic modulus and graded deformation under tensile tests. Regarding property closure foundations, prior work established linear, symmetry-aware orientation distribution function (ODF)-based texture-property linkages of metallic alloys for efficient mapping of elastic coefficients and microstructure design [47–50]. Acar and Sundararaghavan [51] developed a linear solution treating nodal ODFs as variables under constraints to achieve target properties, later extended to reliability-based design with uncertainty propagation and probability-of-failure constraints [52]. For efficiency, Eghtesad et al. [53] proposed compact ODF reconstructions using generalized spherical harmonics to preserve texture features, while a data-driven machine learning methods integrate with physics-based closures for exploration of feasible property regions capturing linear and nonlinear mappings [54].

Optimization methods that minimize strain energy have demonstrated exceptional effectiveness in reducing structural deformation and achieving uniform stress distributions [55–57]. By minimizing the total strain energy within a structure, deformation can be significantly reduced, leading to lightweight and mechanically efficient designs [58–60]. Building on these established frameworks, this research introduces a novel structural optimization methodology that incorporates micro- and meso-scale effects into continuum-level design. Inconel 718 (IN-718) is selected as the metallic base material for its microstructural tunability, while polylactic acid (PLA) is chosen for its availability, processability, and broad applicability. Feasible stiffness coefficient spaces are constructed using a constraint build-up method based on grain orientation distributions, while the Malek and Gibson [61] approach is employed to define honeycomb structure property spaces as functions of geometry and density. The effects of feature-level degrees of freedom on property-space morphology are also analyzed, including cross-scale analogies in how property closures are constructed. An iterative strain energy-based objective function minimizes structural deformation while satisfying the defined micro- and meso-scale constraints. The methodology further tracks the evolution of optimal property points and their corresponding microstructure textures and unit cell geometries throughout the optimization process. Finally, comparative analyses between optimized and baseline configurations validate the proposed FGM design strategy. To the authors' knowledge, no prior work has fully coupled unified, scale-agnostic stiffness variations with a continuum scale strain energy framework. This applies regardless of scale, whether variations arise from microscopic grain orientation driven stiffness changes or mesoscopic unit cell

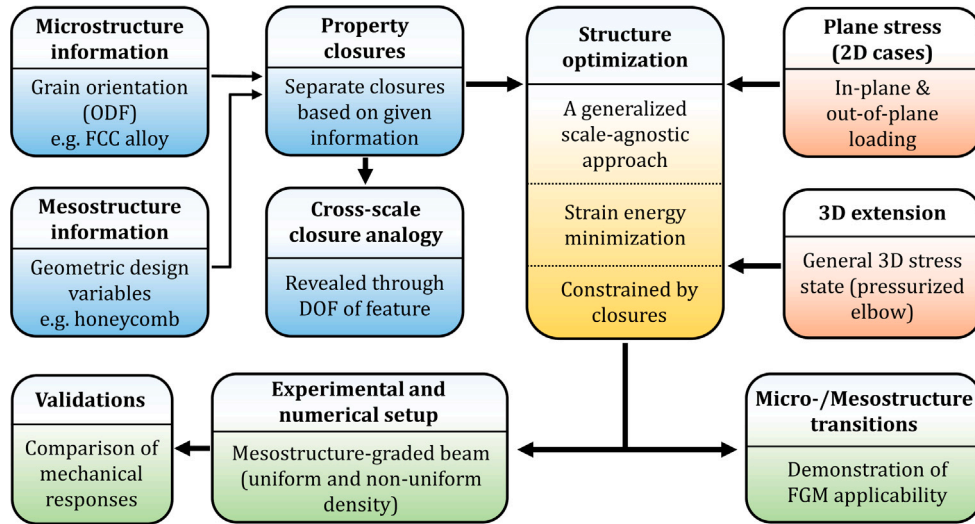


Fig. 1. An overview of the framework for this study, represented by a flow diagram.

driven stiffness changes. Addressing this gap enables controlled, graded material behavior without altering the macroscopic geometry.

The remainder of this paper is organized as follows. Section 2 introduces the proposed methodology, detailing the derivation of micro- and mesostructure-informed property closures (as mathematical constraints), the finite element framework for both in-plane and out-of-plane loading, and the iterative strain energy-based optimization schemes for element-wise elastic stiffness coefficients. Section 3 illustrates property closures, reveals the effect of design variables on the shape of both micro- and mesostructure-driven property closures, and discusses the results obtained from finite element simulations, highlighting their effectiveness and demonstrating significant improvements over conventional design strategies. Section 4 extends the study by illustrating how the proposed generalized approach can be readily applied to three-dimensional structures. Fig. 1 provides a summary overview of this study, where blue-shaded boxes highlight property closures and constraints, the peach-shaded box covers finite element analysis, the yellow-shaded box represents optimization, and the green-shaded boxes correspond to real-world applicability and validation. Finally, Section 5 summarizes the key findings and outlines potential directions for future research aimed at advancing the practical implementation of this work in engineering applications.

2. Methodology

This section presents the proposed methodology for developing property closures under microstructural texture variations and mesostructural geometry and density variations. It further explains how these design spaces can be described using mathematical constraints, enabling their use in strain energy-based structural optimization to achieve improved structural performance.

2.1. Developing property closure

Variations in grain orientation and meso-scale geometric unit-cell features influence the elastic properties in distinct ways, which necessitates separate methods for constructing their respective property spaces.

2.1.1. Microstructure-informed property closure

The study first examines structures where the plane stress assumption applies, allowing the elastic response to be fully characterized by the in-plane stiffness coefficients C_{11} , C_{22} , C_{12} , and C_{66} . This formulation leverages the grain orientation variations within the microstructure for constructing property spaces of the IN-718 alloy. The

crystallographic texture of the IN-718 microstructure is characterized using ODFs. These functions represent the volume densities of different crystallographic orientations within a polycrystalline material and are employed to derive homogenized properties. In this study, the Rodrigues orientation framework based on the axis-angle representation is utilized. Within this formulation, the rotation axis is scaled according to $\mathbf{r} = \mathbf{n} \tan(\varphi/2)$ where $\mathbf{n}$ represents the unit vector defining the rotation axis and φ is the associated rotation angle [62], as illustrated in Fig. 2(a). The relationship between the lattice orientation matrix $\mathbf{R}$ and the Rodrigues rotation vector $\mathbf{r}$ is expressed through Eq. (1) [63]. In this expression, $\mathbf{I}$ denotes the identity tensor, while the scalar $\mathbf{r} \cdot \mathbf{r}$ corresponds to the squared magnitude of the Rodrigues rotation vector $\mathbf{r}$. The term $\mathbf{r} \otimes \mathbf{r}$ represents the dyadic (outer) product of $\mathbf{r}$ with itself, and $\mathbf{I} \times \mathbf{r}$ denotes the skew-symmetric tensor associated with the cross product. The magnitude of each orientation vector is inherently constrained by a maximum value, which corresponds to the highest possible orientation within a symmetric three-dimensional domain. When these vectors are mapped in three-dimensional space, their endpoints form intricate polyhedral structures, illustrated in Fig. 2(b). In polycrystalline materials with an FCC structure, the inherent crystal symmetry simplifies the ODF. Owing to these symmetry constraints, only 76 unique orientations need to be evaluated out of the 145 possible orientations, as illustrated in Fig. 2(c) [47,64].

Subsequently, this local finite element discretization approach is applied within the Rodrigues orientation space framework to evaluate the homogenized properties. Eq. (2) defines the ODFs (A), incorporating the requirement that crystal volume densities conform to a constraint ensuring overall volume normalization [52,65]. The value of the ODF at the m th integration point is denoted by $A(\mathbf{r}_m)$, where $\mathbf{r}_m$ is the corresponding global coordinate within the n th finite element. The determinant of the Jacobian for this element is given by $|J_n|$, and w_m represents the associated quadrature weight. The computational framework comprises N distinct nodes and N_{elm} finite elements, each containing N_{int} integration points. Each nodal ODF value corresponds to the volume density of its associated crystallographic orientation. The homogenized material property, denoted by $\langle \chi \rangle$, is determined by integrating the orientation-dependent single-crystal property values (χ) with the nodal ODFs, as expressed in Eq. (3). This analysis employs the single-crystal stiffness matrix of IN-718 provided in Ref. [66], with constants $C_{11} = 242.18$ GPa, $C_{12} = 138.85$ GPa, and $C_{66} = 104.20$ GPa.

$$\mathbf{R} = \frac{1}{1 + \mathbf{r} \cdot \mathbf{r}} (\mathbf{I} (1 - \mathbf{r} \cdot \mathbf{r}) + 2(\mathbf{r} \otimes \mathbf{r} + \mathbf{I} \times \mathbf{r})). \quad (1)$$

$$\int_{\Omega} A(\mathbf{r}) d\mathbf{v} = \sum_{n=1}^{N_{\text{elm}}} \sum_{m=1}^{N_{\text{int}}} A(\mathbf{r}_m) w_m |J_n| \frac{1}{(1 + \mathbf{r}_m \cdot \mathbf{r}_m)^2} = 1. \quad (2)$$

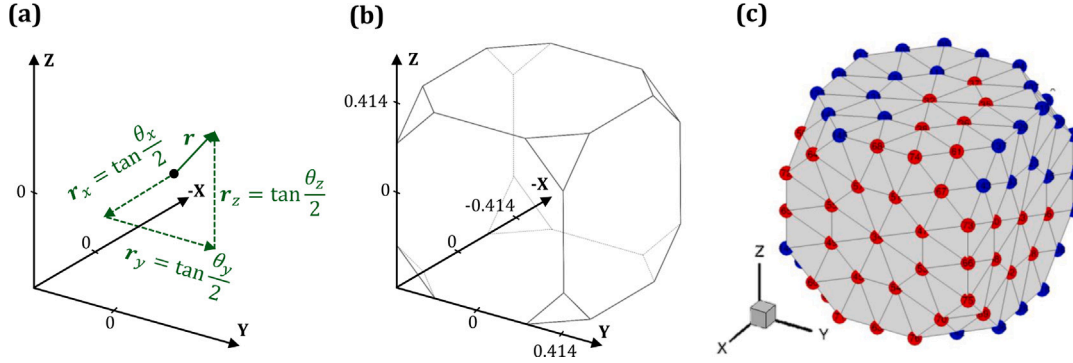


Fig. 2. Illustration of fundamental concepts within the Rodrigues orientation framework: (a) The correlation between the Rodrigues vector ($\mathbf{r}$) and the angular parameters defining crystal orientation. (b) The fundamental space of the Rodrigues space corresponding to FCC crystal symmetry. (c) A finite element discretization of the Rodrigues space, in which ODF are assigned to the mesh nodes. Independently defined ODFs are indicated in red, whereas dependent ODFs are represented in blue.

$$\langle \chi \rangle = \int_{\Omega} \chi(\mathbf{r}) A(\mathbf{r}) d\mathbf{v} = \sum_{n=1}^{N_{elm}} \sum_{m=1}^{N_{int}} \chi(\mathbf{r}_m) A(\mathbf{r}_m) w_m |J_n| \frac{1}{(1 + \mathbf{r}_m \cdot \mathbf{r}_m)^2}. \quad (3)$$

The normalization requirement can be equivalently represented as the linear relation $\mathbf{Q}^T \mathbf{A} = 1$, where the expression is defined in terms of the ODF values evaluated at the nodal points. Then, a numerical approach is formulated to determine the property closure which represents a convex space encompassing all attainable values of selected stiffness properties through feasible microstructure configurations. To build this property space for FCC IN-718 based on grain orientations, the extreme orientations, denoted by the vertex texture ($\mathbf{A}_v$) formulation, are initially identified via Eq. (4). Next, the intermediate orientations ($\mathbf{A}_{inter}$) are produced by interpolating between vertex pairs, following Eq. (5). The unified collection of $\mathbf{A}_v$ and $\mathbf{A}_{inter}$ establishes the full property space associated with a specified set of in-plane elastic stiffness components. Both Eqs. (4) and (5) are designed to satisfy the statistical volume normalization constraint, and the interpolation points are distributed according to a weight parameter w , which ensures a linear spacing between 0 and 1. To generalize the interpolation to an arbitrary number n of vertex points $\mathbf{A}_v^{(k)}$ (for $k = 1, 2, \dots, n$), use a convex combination with weights w_k that sum to 1 and are non-negative, as shown in Eq. (6). Thus, the microstructure-informed property space should contain all property points associated with any possible values of $\mathbf{A}_{inter}$ obtained from Eq. (6).

$$\mathbf{A}_v^{(i)} = [0, \dots, \frac{1}{Q_i}, \dots, 0]^T, \quad (\text{only the } i\text{th entry is } \frac{1}{Q_i}). \quad (4)$$

$$\mathbf{A}_{inter} = \frac{w \mathbf{A}_v^{(i)} + (1-w) \mathbf{A}_v^{(j)}}{\mathbf{Q}^T \cdot (w \mathbf{A}_v^{(i)} + (1-w) \mathbf{A}_v^{(j)})}, \quad \text{where } w \in [0, 1]. \quad (5)$$

$$\mathbf{A}_{inter} = \frac{w_1 \mathbf{A}_v^{(1)} + w_2 \mathbf{A}_v^{(2)} + \dots + w_{n-1} \mathbf{A}_v^{(n-1)} + w_n \mathbf{A}_v^{(n)}}{\mathbf{Q}^T \cdot (w_1 \mathbf{A}_v^{(1)} + w_2 \mathbf{A}_v^{(2)} + \dots + w_{n-1} \mathbf{A}_v^{(n-1)} + w_n \mathbf{A}_v^{(n)})}, \quad (6)$$

where $w_1 + w_2 + \dots + w_{n-1} + w_n = 1$, and $w_k \geq 0 \forall k$.

In this work, we first selected C_{11} , C_{22} , and C_{12} to develop constraint boundaries derived from their respective closures. Given the inherent linear relationship between C_{12} and C_{66} , an additional equality constraint is introduced to capture this dependency. The extreme points are derived from the vertices and the interpolation property points, which are obtained using Eqs. (4) and (5), respectively. The closure or convex hull is determined using a Delaunay triangulation-based algorithm, yielding a set of triangular facets that enclose the feasible region. Each facet is defined by three points $\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3$, and the plane equation for the facet is derived as follows:

- (i) Compute two edge vectors: $\mathbf{v}_1 = \mathbf{p}_2 - \mathbf{p}_1$, $\mathbf{v}_2 = \mathbf{p}_3 - \mathbf{p}_1$, spanning the plane of the facet.

- (ii) Calculate the normal vector: $\mathbf{n} = \frac{\mathbf{v}_1 \times \mathbf{v}_2}{\|\mathbf{v}_1 \times \mathbf{v}_2\|}$, normalized to unit length which provides a consistent direction perpendicular to the facet.
- (iii) Determine the plane equation: $\mathbf{n} \cdot [C_{11}, C_{22}, C_{12}] = d$, where constant $d = \mathbf{n} \cdot \mathbf{p}_1$.
- (iv) Formulate the inequality: $\mathbf{n} \cdot [C_{11}, C_{22}, C_{12}] \leq d$ (or $\geq d$, depending on the normal direction) to ensure the test point (isotropic point where the grains are randomly oriented) lies on the correct side of the plane, constraining the inequality constraints of the property closure.

2.1.2. Mesosstructure-informed closure

Honeycomb-core sandwich panels, as illustrated in Fig. 3(a), have long been recognized as one of the efficient structural configurations for achieving high bending stiffness and strength at minimal weight [67–70]. The thin face sheets carry in-plane and bending stresses, whereas the honeycomb core, characterized by its exceptionally high out-of-plane specific shear modulus, separates the faces and resists transverse shear, thereby greatly increasing flexural rigidity and delaying local face wrinkling. The cellular architecture further enhances buckling resistance, improves energy absorption under impact, and contributes to damping, thermal, and acoustic performance. Hexagonal honeycombs are inherently anisotropic: their mechanical properties differ along the ribbon and expansion directions, with the highest stiffness occurring through the out-of-plane thickness direction [71]. Consequently, the core is conventionally placed vertically within the sandwich beam; however, the remaining two in-plane directions remain relatively weak, which may represent a potential design limitation of such structures [72,73]. To address this issue, the present study incorporates optimization of the honeycomb's geometric parameters to enhance its performance under loads applied along either the ribbon or expansion directions. Accordingly, Fig. 3(b) presents the configuration of the simply supported beam, in which the honeycomb features are aligned along the loading face, the strong direction is oriented in the out-of-plane direction, and the in-plane unit-cell geometric parameters are defined by the wall thickness (t), effective bending lengths (l_b, h_b), cell angle (θ), and wall aspect ratio (h/l).

In order to capture the local responses of the honeycomb unit cell, the homogenization method is adopted to characterize the properties of its underlying elements. In the present study, the analytical framework proposed by Malek and Gibson [61] is employed, as it has been validated through both experimental investigations and finite element analyses. This formulation is subsequently utilized to construct the property space of the in-plane stiffness components, subject to realistic geometric and relative density constraints. This analytical model refines the classical Gibson and Ashby [74] approach by explicitly incorporating the influence of nodal regions where inclined and vertical cell

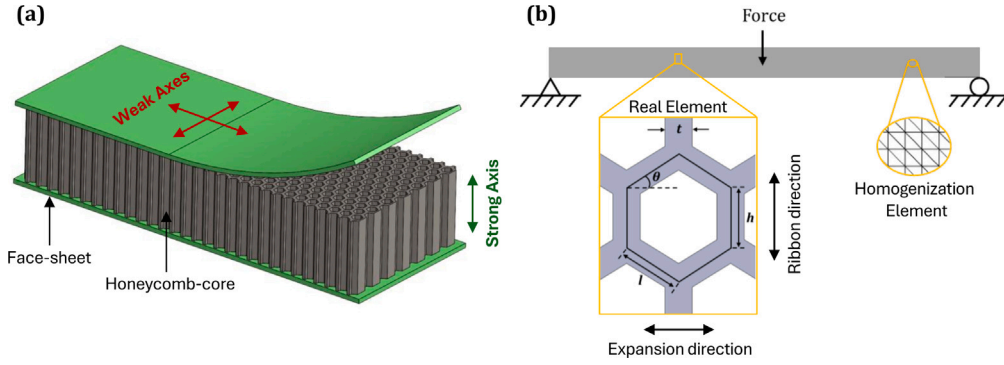


Fig. 3. (a) Conventional application of a honeycomb lattice core structure, highlighting its fundamental features; (b) schematic illustration of the simply supported beam considered for optimizing the underlying mesostructured honeycomb unit cell (the real element represents the actual element with consideration of the porous honeycomb structure, while the homogenized element represents an equivalent solid element in which the porous structure is idealized as a perfect solid).

walls intersect. The model introduces effective bending lengths to represent local bending and shear effects, thereby improving the prediction accuracy of elastic responses at moderate to high relative densities. Closed-form expressions are established for both in-plane elastic moduli and out-of-plane shear components, extending the analytical scope beyond the traditional bounds available in earlier formulations. The application of this analytical approach for constructing the property space is described in Appendix A. Another instance of the double-arrowhead mesostructure is examined to develop and analyze the corresponding property space using the approach proposed by Berinskii [75] and Huber's method [76], which has been added to the 'Supplementary Materials'.

2.2. FEM framework

Fig. 4 contains a flowchart which illustrates an iterative FEM framework designed to optimize orthotropic material properties for structural components by minimizing strain energy. The framework begins with the initialization of the mesh, applied loads, and boundary conditions, followed by the assignment of initial element-wise material stiffness properties, C . Using these properties, the element stiffness matrices K_e are constructed and assembled into a global system to evaluate displacements across the structure. From the computed displacements, element strains ϵ and stresses σ are derived, which are then used to formulate a strain energy function. This function serves as the basis for optimizing C for each element using numerical minimization techniques to reduce the overall strain energy. A convergence check is performed after each iteration to determine if the solution has stabilized; if not, the updated material properties are used for the next iteration.

This study develops the FEM frameworks for analyzing the two-dimensional structures with orthotropic material properties as mentioned earlier, integrated with an iterative optimization scheme to tailor element-specific stiffness coefficients. This encompasses two main components: the FEM formulation and the material property optimization. Two different plane stress scenarios have been analyzed: one involving in-plane loading in beam and frame structures, and the other involving out-of-plane loading in a plate. Although the FEM formulation schemes differ for the two loading cases, the underlying property optimization technique remains consistent across both scenarios.

2.2.1. In-plane loading

The FEM formulation discretizes the 2D domain using a Delaunay triangulation to generate triangular elements. Each node possesses two degrees of freedom (DOFs) in these constant strain triangle (CST) elements, corresponding to displacements in the x - and y -directions (u_x, u_y). The global stiffness matrix K is assembled by computing element stiffness matrices K_e for each triangular element under a plane

stress assumption [77,78]. The element stiffness matrix is formulated as shown in Eq. (7) where t is the element thickness, A is the element area computed using the polygon area method, B is the strain-displacement matrix, and C is the material constitutive matrix. For an orthotropic material under plane stress, C is defined by Eq. (8) where C_{11} , C_{22} , C_{12} , and C_{66} are the stiffness coefficients representing the longitudinal, transverse, coupling, and shear moduli, respectively. The strain-displacement matrix B for a 3-node triangular element is constructed using shape function derivatives as represented in Eq. (9) where $N_{x_i} = \beta_i/(2A)$, $N_{y_i} = \gamma_i/(2A)$, and β_i, γ_i are derived from nodal coordinates as represented in Eq. (9). Appropriate boundary conditions and external loads are applied, and the system is solved for displacements $U = K^{-1}F$ where F is the force vector. Element-wise strains and stresses are computed as $\epsilon_e = Bu_e$ and $\sigma_e = C\epsilon_e$, respectively.

$$K_e = tAB^T C B. \quad (7)$$

$$C = \begin{bmatrix} C_{11} & C_{12} & 0 \\ C_{12} & C_{22} & 0 \\ 0 & 0 & C_{66} \end{bmatrix}. \quad (8)$$

$$B = \begin{bmatrix} N_{x_1} & 0 & N_{x_2} & 0 & N_{x_3} & 0 \\ 0 & N_{y_1} & 0 & N_{y_2} & 0 & N_{y_3} \\ N_{y_1} & N_{x_1} & N_{y_2} & N_{x_2} & N_{y_3} & N_{x_3} \end{bmatrix}. \quad (9)$$

$$\beta = \begin{bmatrix} y_2 - y_3 \\ y_3 - y_1 \\ y_1 - y_2 \end{bmatrix}, \quad \gamma = \begin{bmatrix} x_3 - x_2 \\ x_1 - x_3 \\ x_2 - x_1 \end{bmatrix}. \quad (10)$$

2.2.2. Out-of-plane loading

The stiffness matrix for a sixteen DOFs rectangular thin plate bending element, which is also known as Bogner-Fox-Schmit (BFS) plate element [79], based on Kirchhoff plate theory [80], is formulated for finite element analysis of thin plates under bending. Each of the four nodes has four DOFs: transverse displacement w , rotations about the x and y axes ($w_x = \frac{\partial w}{\partial x}$ and $w_y = \frac{\partial w}{\partial y}$), and twist curvature ($w_{xy} = \frac{\partial^2 w}{\partial x \partial y}$). The element, with dimensions $a \times b$ and thickness t , accounts for orthotropic material properties via stiffness coefficients. The element stiffness matrix $K_e \in \mathbb{R}^{16 \times 16}$ is illustrated in Eq. (11), where K_1 , K_2 , K_3 and K_4 are component matrices derived from shape function derivatives.

$$K_e = \frac{t^3}{12} (C_{11}K_1 + C_{22}K_2 + C_{12}K_3 + 4C_{66}K_4). \quad (11)$$

$$\begin{aligned} K_1(i, j) &= r_{1a}(n_i, n_j) \cdot r_{2b}(m_i, m_j), \\ K_2(i, j) &= r_{2a}(n_i, n_j) \cdot r_{1b}(m_i, m_j), \\ K_3(i, j) &= s_{1a}(n_i, n_j) \cdot s_{1b}(m_j, m_i) + s_{1a}(n_j, n_i) \cdot s_{1b}(m_i, m_j), \\ K_4(i, j) &= r_{3a}(n_i, n_j) \cdot r_{3b}(m_i, m_j). \end{aligned} \quad (12)$$

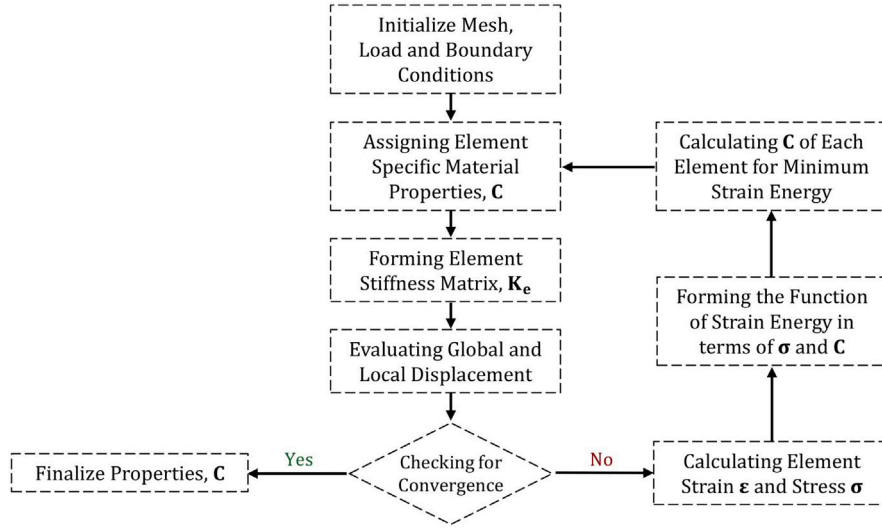


Fig. 4. Flowchart illustrating the iterative FEM process used to optimize the orthotropic material properties of the underlying elements. This generalized framework is applied to both micro- and mesostructural grading.

The constituent matrices are defined in Eq. (12), where the indices $i, j = 1, \dots, 16$, and $n_i, m_i \in \{1, 2, 3, 4\}$ are indexing vectors that map each global DOF to subcomponents of the matrix blocks. The component matrices are assembled using index vectors: $\mathbf{n} = [1, 3, 1, 3, 1, 3, 1, 3, 2, 4, 2, 4, 2, 4, 2, 4]$ and $\mathbf{m} = [1, 1, 3, 3, 2, 2, 4, 4, 2, 2, 4, 4, 1, 1, 3, 3]$. The matrices $\mathbf{r}_{1a}, \mathbf{r}_{2a}, \mathbf{r}_{3a}, \mathbf{s}_{1a}$ and their counterparts in the b -direction are defined from shape function derivatives and integration over the element domain. For the x -direction (a), sub-matrices $\mathbf{r}_{1a}, \mathbf{r}_{2a}, \mathbf{r}_{3a}, \mathbf{s}_{1a}$ are expressed in Eq. (13). For the y -direction (b), sub-matrices $\mathbf{r}_{1b}, \mathbf{r}_{2b}, \mathbf{r}_{3b}, \mathbf{s}_{1b}$ are defined similarly, replacing a with b [81,82].

$$\mathbf{r}_{1a} = \begin{bmatrix} \frac{12}{a^3} & -\frac{12}{a^3} & \frac{6}{a^2} & -\frac{6}{a^2} \\ -\frac{12}{a^3} & \frac{12}{a^3} & -\frac{6}{a^2} & \frac{6}{a^2} \\ \frac{6}{a^2} & -\frac{6}{a^2} & \frac{4}{a} & -\frac{4}{a} \\ \frac{6}{a^2} & -\frac{6}{a^2} & -\frac{4}{a} & \frac{4}{a} \end{bmatrix}, \quad \mathbf{r}_{2a} = \begin{bmatrix} \frac{13a}{35} & \frac{9a}{70} & \frac{11a^2}{210} & -\frac{13a^2}{420} \\ \frac{9a}{70} & \frac{13a}{35} & \frac{13a^2}{420} & -\frac{11a^2}{210} \\ \frac{11a^2}{210} & \frac{13a^2}{420} & \frac{a^3}{105} & -\frac{a^3}{140} \\ -\frac{13a^2}{420} & -\frac{11a^2}{210} & -\frac{a^3}{140} & \frac{a^3}{105} \end{bmatrix},$$

$$\mathbf{r}_{3a} = \begin{bmatrix} \frac{6}{5a} & -\frac{6}{5a} & \frac{1}{10} & -\frac{1}{10} \\ -\frac{6}{5a} & \frac{6}{5a} & -\frac{1}{10} & \frac{1}{10} \\ \frac{1}{10} & -\frac{1}{10} & \frac{2a}{15} & -\frac{a}{30} \\ \frac{1}{10} & -\frac{1}{10} & -\frac{a}{30} & \frac{2a}{15} \end{bmatrix}, \quad \mathbf{s}_{1a} = \begin{bmatrix} -\frac{6}{5a} & \frac{6}{5a} & -\frac{1}{10} & \frac{1}{10} \\ \frac{6}{5a} & -\frac{6}{5a} & \frac{1}{10} & -\frac{1}{10} \\ -\frac{1}{10} & \frac{1}{10} & -\frac{2a}{15} & \frac{a}{30} \\ -\frac{1}{10} & \frac{1}{10} & \frac{a}{30} & -\frac{2a}{15} \end{bmatrix}. \quad (13)$$

Later, appropriate boundary conditions and external loads are applied, and the system is solved for displacements $\mathbf{W} = \mathbf{K}^{-1}\mathbf{F}$ where $\mathbf{F}$ is the force vector. Strains are computed based on the curvature of the plate, derived from the second derivatives of the transverse displacement $w(x, y)$ as depicted in Eq. (14). According to Kirchhoff plate theory, the strains at a distance z from the midplane (neutral axis) are related to the curvatures of the plate. The strains at the top surface ($z = t/2$) are considered. Second derivatives of Eq. (14) are approximated using central differences for interior nodes and the mixed derivative $w_{xy} = \frac{\partial^2 w}{\partial x \partial y}$ is directly taken from the DOF vector which has been solved early. Thereafter, the element-averaged strains are computed by averaging nodal strains over the four nodes of each element. Subsequently, element specific stresses are calculated using the constitutive relationship for an orthotropic material, relating strains to stresses via the stiffness matrix. The element-specific stiffness coefficients are

utilized to compute stresses for each element ($\sigma_e = \mathbf{C}_e \epsilon_e$).

$$\epsilon_e = \begin{bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \gamma_{12} \end{bmatrix} = -z \boldsymbol{\kappa} = \begin{bmatrix} -z \frac{\partial^2 w}{\partial x^2} \\ -z \frac{\partial^2 w}{\partial y^2} \\ -2z \frac{\partial^2 w}{\partial x \partial y} \end{bmatrix}. \quad (14)$$

2.3. Optimization of elastic stiffness

The optimization process aims to minimize the total strain energy by optimizing the elastic stiffness coefficients C_{11}, C_{22}, C_{12} , and C_{66} for each element over multiple iterations. The objective function (f) is formulated to reduce deformation, which correlates with minimizing the strain energy. For each element in all iterations, the objective function is defined using the original definition of strain energy, as presented in Eq. (15), where U_e denotes the strain energy stored within the element, and the total strain energy is obtained by summing over all elements. Subsequently, the objective function is simplified as shown in Eq. (16), where $\sigma_{11}, \sigma_{22}, \tau_{12}$ represent the stress components computed from the previous iteration. This function indirectly minimizes the strain energy by optimizing the material properties to reduce strain magnitudes. The optimization is constrained by the inequality constraints ($\mathbf{A}_{\text{ineq}} \mathbf{x} \leq \mathbf{b}_{\text{ineq}}$) and the equality constraints ($\mathbf{A}_{\text{eq}} \mathbf{x} = \mathbf{b}_{\text{eq}}$) depending upon the constraints, where $\mathbf{x} = \{C_{11}, C_{22}, C_{12}, C_{66}\}$ as mentioned previously.

$$U_e = \frac{1}{2} \int_V \boldsymbol{\sigma}^T \boldsymbol{\epsilon} dV = \frac{1}{2} t \mathbf{A}_e^T \mathbf{C}_e \boldsymbol{\epsilon}_e. \quad (15)$$

$$f(\mathbf{C}) = \left(\frac{-\sigma_{11}^2 C_{22} + 2\sigma_{11}\sigma_{22}C_{12} - \sigma_{22}^2 C_{11}}{C_{12}^2 - C_{11}C_{22}} \right) + \frac{\tau_{12}^2}{C_{66}}. \quad (16)$$

The initial guess for the first iteration uses the uniform stiffness coefficients (corresponding to the isotropic property point), while subsequent iterations use the optimized values from the prior iteration. The simplified objective function proportional to strain energy is expressed in terms of stress components derived from the previous iteration. The interior-point optimization algorithm [83,84] solves the constrained nonlinear optimization problem. Optimized stiffness coefficients are stored and used in the FEM calculations with element-specific properties to compute updated stresses, strains, and displacements.

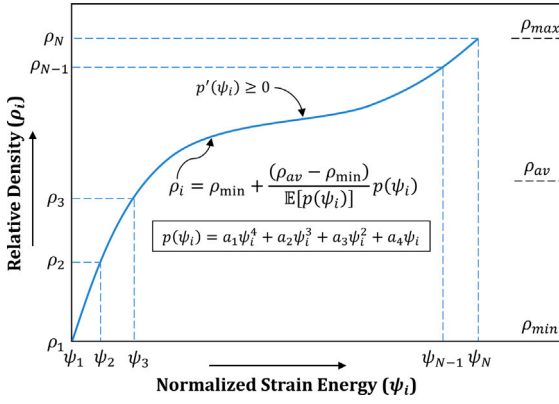


Fig. 5. Illustration of the schematic showing the relative density distribution of the unit cells corresponding to the FEM elements plotted against the normalized strain energy, with several characteristic features of the curve indicated.

2.4. Optimization of local density distribution

In the iterative topology optimization framework for the honeycomb unit cell structured beam, the local density ρ for each finite element is dynamically adjusted based on the strain energy density from the previous FEM iteration. This adjustment aims to concentrate material in high-strain energy regions while adhering to a global average density constraint ρ_{av} , within a bounded range of $\rho_{\min}$ to $\rho_{\max}$. Fig. 5 illustrates the fundamental concept of the relative density distribution in FEM elements, showing how strain energy influences the determination of relative density. For the i th element among total of N , the target density ρ_i is given by Eq. (17) where $\psi_i = U_{ei}/\bar{U}_e$ is the normalized energy density (where $\bar{U}_e$ is the spatial average), $p(\psi_i)$ is the quartic evaluation, and $\mathbb{E}[p(\psi_i)]$ is the spatial average of $p(\psi_i)$ over all elements. The density mapping employs a mean-preserving quartic polynomial, whose coefficients are pre-optimized via a genetic algorithm (GA) to minimize structural compliance, ensuring an efficient starting point for the multi-iteration process. The coefficients a_1 , a_2 , a_3 , and a_4 for the quartic mapping $p(\psi_i) = a_1\psi_i^4 + a_2\psi_i^3 + a_3\psi_i^2 + a_4\psi_i$ are determined using a dedicated optimization routine on a reduced mesh to accelerate computation.

$$\rho_i = \rho_{\min} + \frac{(\rho_{av} - \rho_{\min})}{\mathbb{E}[p(\psi_i)]} p(\psi_i). \quad (17)$$

This routine begins with an initial uniform-material FEM to compute the baseline strain energy density as represented in Eq. (15), normalized element-wise as ψ_i . The GA minimizes the total integrated strain energy (compliance) after a single density-mapping and material-selection step, subject to monotonicity constraints ensuring $p'(\psi_i) \geq 0$ for ψ_i in the relevant range (estimated from a high percentile of normalized energies) and $a_4 \geq 0$. These constraints are enforced via linear inequalities evaluated at discrete points, preventing non-physical density allocations. The GA operates with a proper population size over a sufficient generations, with bounds for each coefficient and an initial guess favoring a near-linear form. The optimized coefficients are then fixed in the main iterative loop, which runs on a finer mesh for several optimization steps post-initial uniform FEM. In each iteration, the quartic mapping redistributes densities based on updated energy fields. This is followed by candidate selection and FEA, progressively refining the density distribution to reduce deflections and compliance. This GA-preoptimized hybrid approach mapping with iterative refinement conserves the total material usage or average density of the structure

as shown in Eq. (18).

$$\begin{aligned} \bar{\rho} &= \frac{1}{N} \sum_{i=1}^N \left[\rho_{\min} + \frac{(\rho_{av} - \rho_{\min})}{\mathbb{E}[p(\psi_i)]} p(\psi_i) \right] \\ &= \rho_{\min} + (\rho_{av} - \rho_{\min}) \frac{1}{N} \sum_{i=1}^N \frac{p(\psi_i)}{\mathbb{E}[p(\psi_i)]} = \rho_{av}. \end{aligned} \quad (18)$$

3. Results

This section presents and explains the derived elastic stiffness closures, highlighting the fundamental similarities between the microstructural and mesostructural information used to construct these closures. It then introduces the optimal structures obtained through a generalized optimization method, which are further validated using numerical simulations and experimental approaches.

3.1. Illustration of property closures

This work first investigates two-dimensional structures operating under plane stress conditions, encompassing both in-plane and out-of-plane loading, where the elastic behavior is characterized by the in-plane stiffness coefficients C_{11} , C_{22} , C_{12} , and C_{66} .

3.1.1. Microstructure-informed property closure

Variations in microstructural texture, primarily due to grain orientation, induce anisotropy in material properties, which is systematically leveraged to construct the property closure. The closure surfaces of C_{11} , C_{22} , and C_{12} corresponding to all inequality constraints are illustrated in cyan color in Fig. 6. Constraint boundaries are formulated based on the closure surfaces, with an additional equality constraint incorporated to account for the intrinsic linear relationship between C_{12} and C_{66} . A total of sixteen inequality constraints defining the feasible space among C_{11} , C_{22} , and C_{12} , along with one equality constraint representing the linear relation between C_{12} and C_{66} , are provided in the 'Supplementary Materials' as Equations (S-1) and (S-2), respectively. Fig. 6(a) presents the property points, which include the corresponding properties of all 76 nodal points of Rodrigues orientation space (referred to as vertex points, A_v), as well as the interpolation points located between two vertices (denoted as A_{inter}). To encompass all possible orientation-dependent properties, the corresponding convex hull surfaces are generated and visualized in transparent cyan. However, few extreme microstructural orientations, denoted by red vertex points in Fig. 6(b), serve as the foundational data for generating the property space using a Delaunay triangulation-based convex hull algorithm. On the other hand, the star-marked point represents the property point corresponding to a randomly oriented microstructure, characterized by an equal probability distribution over all grain orientations. This isotropic point is incorporated into the construction of the inequality constraints, serving as a reference to ensure that the convex hull properly bounds the feasible property space, as previously described. Each triangular facet of the resulting hull defines a plane equation and a corresponding inequality constraint, ensuring that all admissible material points lie within the physically realizable region of the constructed property space.

3.1.2. Mesostructure-informed property closure

To construct the mesostructure-informed property space, PLA is selected as the base material, as we have the experimental capability to process PLA. First, ASTM D638 tensile specimens are additively manufactured using commercial-grade PLA. These samples are tested on a universal testing machine to determine the isotropic Young's modulus and Poisson's ratio. In the first phase, the complete stress-strain curves are obtained using a single extensometer capable of capturing large deformations, as illustrated in Fig. 7(a). In the second

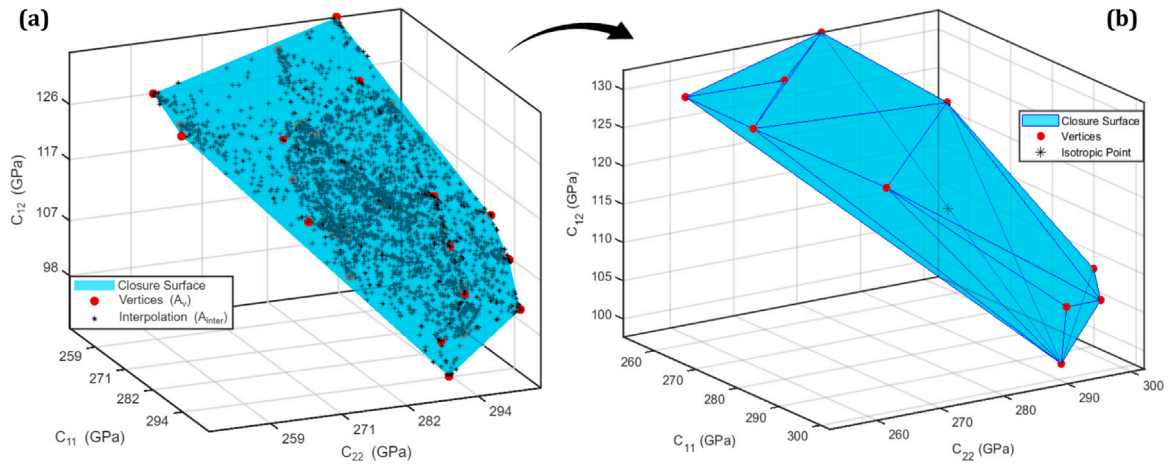


Fig. 6. Illustration of microstructure-informed property closure for stiffness coefficients: (a) closure includes all vertex points (A_v) and all interpolation points (A_{inter}), (b) a simplified representation of (a) showing only the constraint surfaces with closure vertices and the isotropic point (obtained from a randomly oriented microstructure).

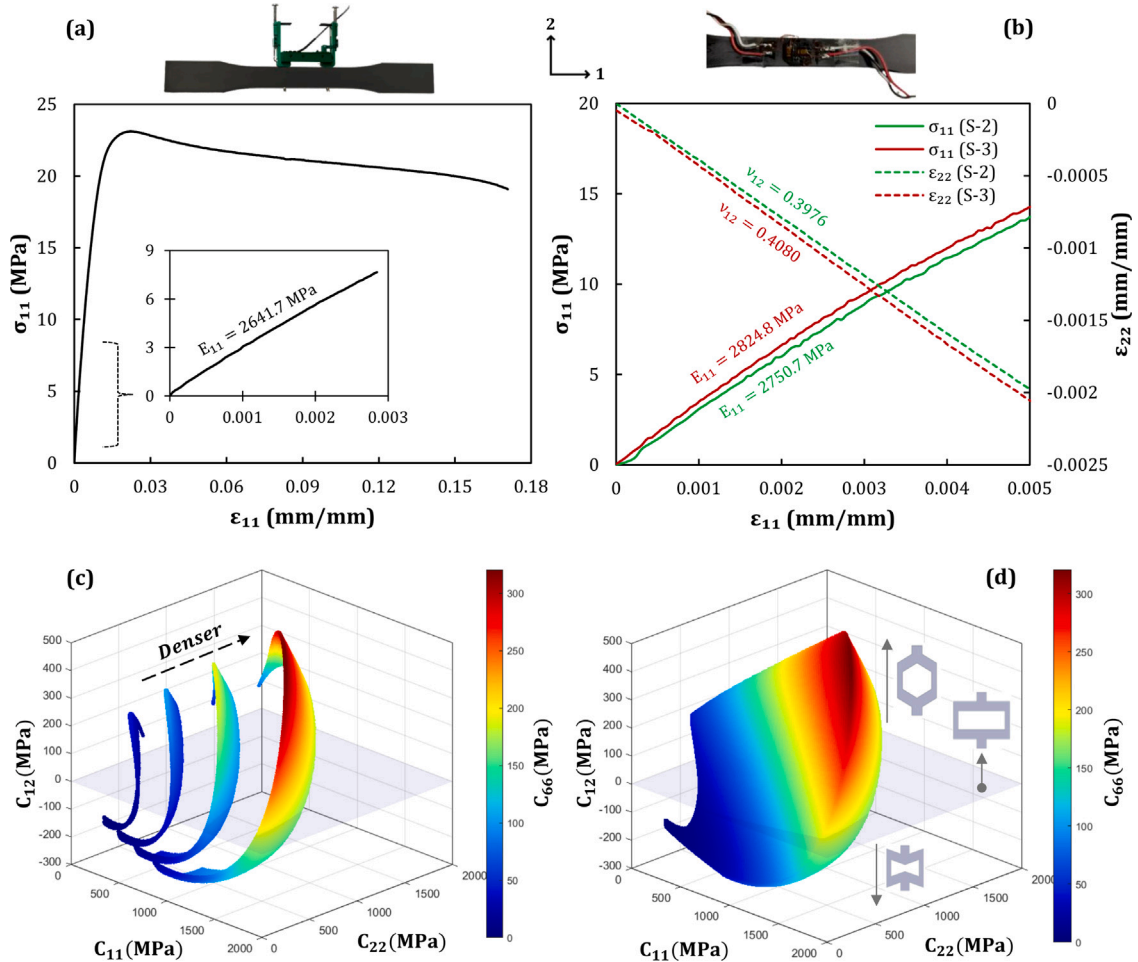


Fig. 7. Illustration of additively manufactured PLA's stress-strain response and mesostructure-informed property space: (a) full stress-strain curve determined using an extensometer; (b) stress-strain curve within the short deformation range determined using longitudinal and transverse strain gauges; (c) property space of honeycomb unit cell for relative densities of 30%, 40%, 50%, and 60%; (d) property space for honeycomb unit cell of continuously varying relative densities from 30% to 60%.

phase, small-deformation stress-strain behavior, along with longitudinal and transverse strain responses, is recorded using strain gauges, as shown in Fig. 7(b). Both experiments come up with the Young's modulus, while the latter additionally provides the Poisson's ratio.

After averaging the results, the Young's modulus and Poisson's ratio are determined as 2739.06 MPa and 0.4028, respectively. These values are subsequently used to construct the property space considering the honeycomb mesostructure.

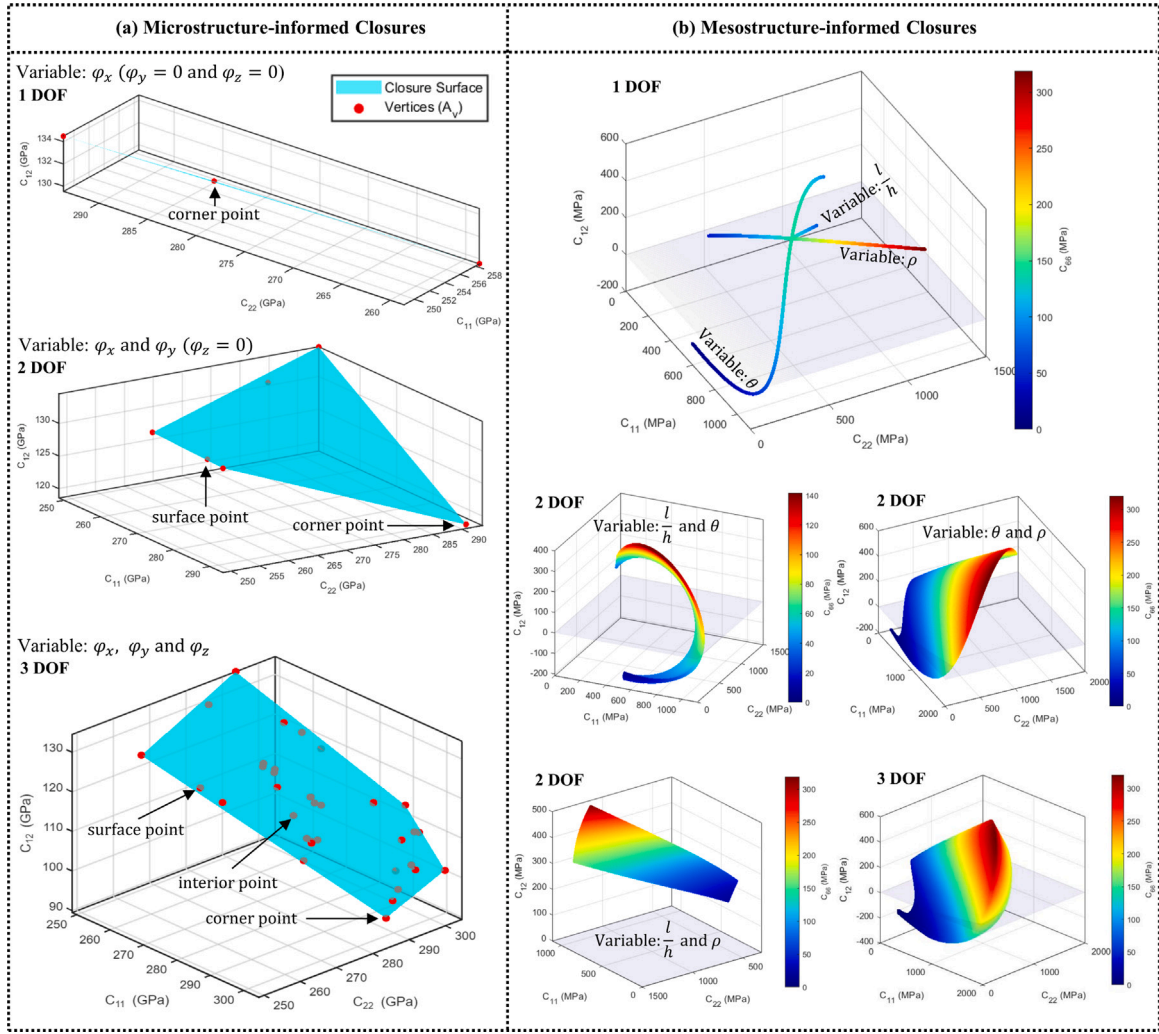


Fig. 8. Representation of the effect of the number of variables (DOF) on the shape of property closures: (a) Microstructure-informed closures where the grain orientation angles serve as the degree of freedom; (b) Mesostructure-informed closures where the geometric parameters of the honeycomb serve as the degree of freedom.

Fig. 7(c) and (d) show the mesostructure-informed property spaces between C_{11} , C_{22} , and C_{12} . Fig. 7(c) represents the property space for relative densities ranging from 30% to 60% with 10% intervals, while Fig. 7(d) presents a continuous density variation within the same range. The overall observation is that, under the consideration of fixed density, the property space forms a smooth surface, whereas continuous density variation produces a non-convex three-dimensional property space. Moreover, denser mesostructures exhibit a wider range of mechanical properties, as the effective surface area increases with density. To ensure physical boundaries and theoretical validity, geometric constraints of the honeycomb are applied as $l/h = [0.5, 1]$, $\theta = [-25^\circ, 60^\circ]$, and $\rho = [0.3, 0.6]$, in accordance with the geometric definition in Fig. 3(b). The black transparent plane in both Fig. 7(c) and (d) contains the rectangular honeycomb properties and distinguishes between regular honeycombs (non-auxetic, $\theta > 0$) and re-entrant honeycombs (auxetic, $\theta < 0$). In the case of a neutral or rectangular honeycomb, C_{12} approaches zero, which results in greater tensile stiffness in C_{11} and C_{22} . Conversely, maximum shear stiffness (C_{66} or G_{12}) is obtained in the case of a perfect hexagonal honeycomb ($\theta = 60^\circ$), as illustrated in property closures with dark red color.

3.1.3. Influence of design variables on the shape of property closures

This subsection aims to reveal the effect of design variables on the shape of both microstructure- and mesostructure-driven property

spaces. In addition, a few common behaviors observed in both types of closures are outlined, which fundamentally arise from similar underlying reasons. In this study, the number of independent design variables is considered as the degree of freedom (DOF). For example, in the microstructure case, the three grain orientation angles are denoted as φ_x , φ_y , and φ_z , while in the mesostructure case, the three geometric parameters of the honeycomb unit cell are represented by l/h , θ , and ρ . Therefore, both cases include a maximum of three degrees of freedom.

The Rodrigues orientation space contains 76 nodal points that serve as the independent texture descriptors (denoted as A_v) representing the grain orientations of the FCC microstructure. These nodal points primarily determine the shape of the property space. Considering all possible grain orientations (3 DOF), a convex three-dimensional property space is exhibited, as described in the previous subsection. Therefore, only the vertex points (A_v) are considered in this subsection to describe the microstructure-driven property spaces. However, reducing the number of DOF can yield different property space shapes, as illustrated in Fig. 8(a). Specifically, when only φ_x (with φ_y and φ_z set to zero) is treated as the design variable, the space consists mainly of corner points, meaning that a set of lines can fully define the closure. When both φ_x and φ_y (with φ_z set to zero) are considered as design variables, the space contains corner points as well as intermediate points located on the hull surface, indicating that a set of surfaces can define the closure. Finally, when all orientation angles are considered, the closure

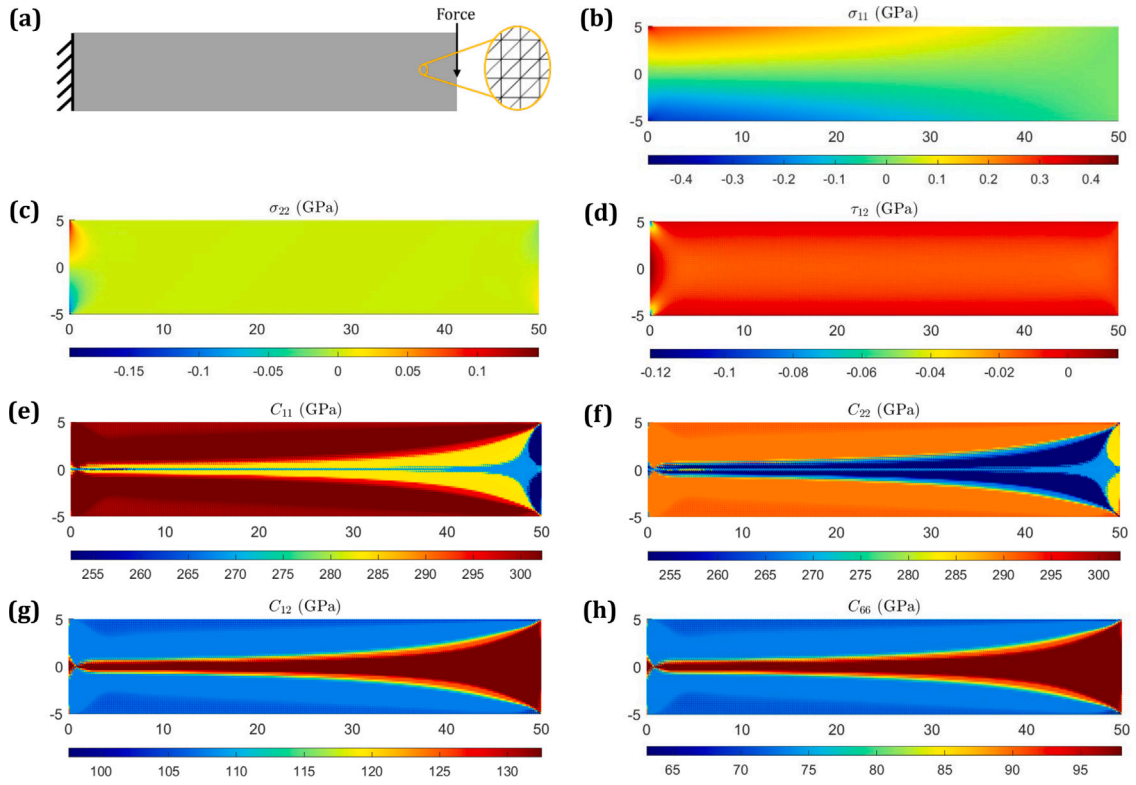


Fig. 9. (a) Schematic diagram of the analyzed cantilever beam subjected to an end load, along with the mesh discretization; and corresponding stress component distributions over the beam at the initial iteration: (b) σ_{11} , (c) σ_{22} , and (d) τ_{12} . The finalized distributions of optimal properties across the cantilever beam after the 4th iteration: (e) C_{11} , (f) C_{22} , (g) C_{12} , and (h) C_{66} . Dimensions are in ‘mm’, and stresses and stiffness coefficients are in ‘GPa’.

includes corner points, surface points, and interior points located inside of the hull surface, the property space becomes a convex 3D domain defined by several inequality constraints, as discussed earlier.

In the case of the honeycomb unit-cell mesostructure-informed property space, the geometric variables are constrained as $l/h = [0.5, 1]$, $\theta = [-25^\circ, 60^\circ]$, and $\rho = [0.3, 0.6]$. When these parameters are held constant, their representative values are taken as $l/h = 1$, $\theta = 30^\circ$, and $\rho = 0.45$. To investigate the influence of these variables on the shape of the property space, reduced-DOF conditions have been considered in a manner analogous to the microstructure case. Detailed observations from Fig. 8(b) reveal that the property space for the 1 DOF condition resembles a curved line, that for the 2 DOF condition exhibits a curved surface, and that for the 3 DOF condition resembles a curved three-dimensional solid domain.

Thus, these overall observations reveal that the shape of property-closure formation fundamentally depends on the number of variables, regardless of the meso- or microstructural consideration. As the DOF increase, they collectively generate a continuum of spaces that ultimately form a solid region in higher-dimensional DOF. However, in our present study, if one additional DOF is introduced (such as the rotation of the micro- or mesostructure) the property points begin to overlap. Furthermore, when discussing dissimilarities, the boundaries of microstructure-informed property spaces appear as either lines or planes due to linear interpolation between two vertex points. This implies that an infinite number of nodal points in Rodrigues space could result in curved lines or curved surfaces defining the property-space boundaries, similar to those observed in mesostructure-driven spaces. On the other hand, the mesostructure-driven property spaces are non-convex and often indistinguishable from microstructure-informed property spaces, owing to the realistic physical and geometric constraints applied to the honeycomb unit cell. In ‘Supplementary Materials’,

another instance of the double-arrowhead mesostructure is examined to highlight how property closure depends on the geometric characteristics of the unit cell.

3.2. Optimization of structures

To illustrate representative optimal structures, several fundamental engineering examples have been selected for explanation. Microstructure-graded designs are created with uniform density (UD), whereas mesostructure-graded designs exhibit both uniform and non-uniform density (NUD) distributions.

3.2.1. Microstructure-graded structures

This portion examines multiple representative cases under plane stress conditions, including in-plane loading scenarios (a cantilever beam with end load) and out-of-plane loading cases (a simply supported thin plate under uniform distributed load). ‘Supplementary Materials’ include a simply supported beam with mid-span load, an L-shaped frame with end load, a tapered cantilever beam with end load, and a clamped thin plate with distributed load. In all cases, optimization begins with uniform isotropic stiffness properties assigned to all finite elements, representing randomly oriented grains in the microstructure (star-marked point in Fig. 6(b)): $C_{11} = C_{22} = 283.5271$ GPa, $C_{12} = 117.0405$ GPa, and $C_{66} = 82.5151$ GPa. The resulting in-plane stress components (σ_{11} , σ_{22} , τ_{12}) are computed and used at the elemental level to formulate the strain energy objective function, with stiffness parameters treated as unknowns to minimize total strain energy. Subsequent iterations update these values, causing only minor stress distribution deviations that cannot alter qualitative behavior; thus, only initial iteration stress distributions are presented. All cases employ very fine meshes for accurate capture of localized stress variations

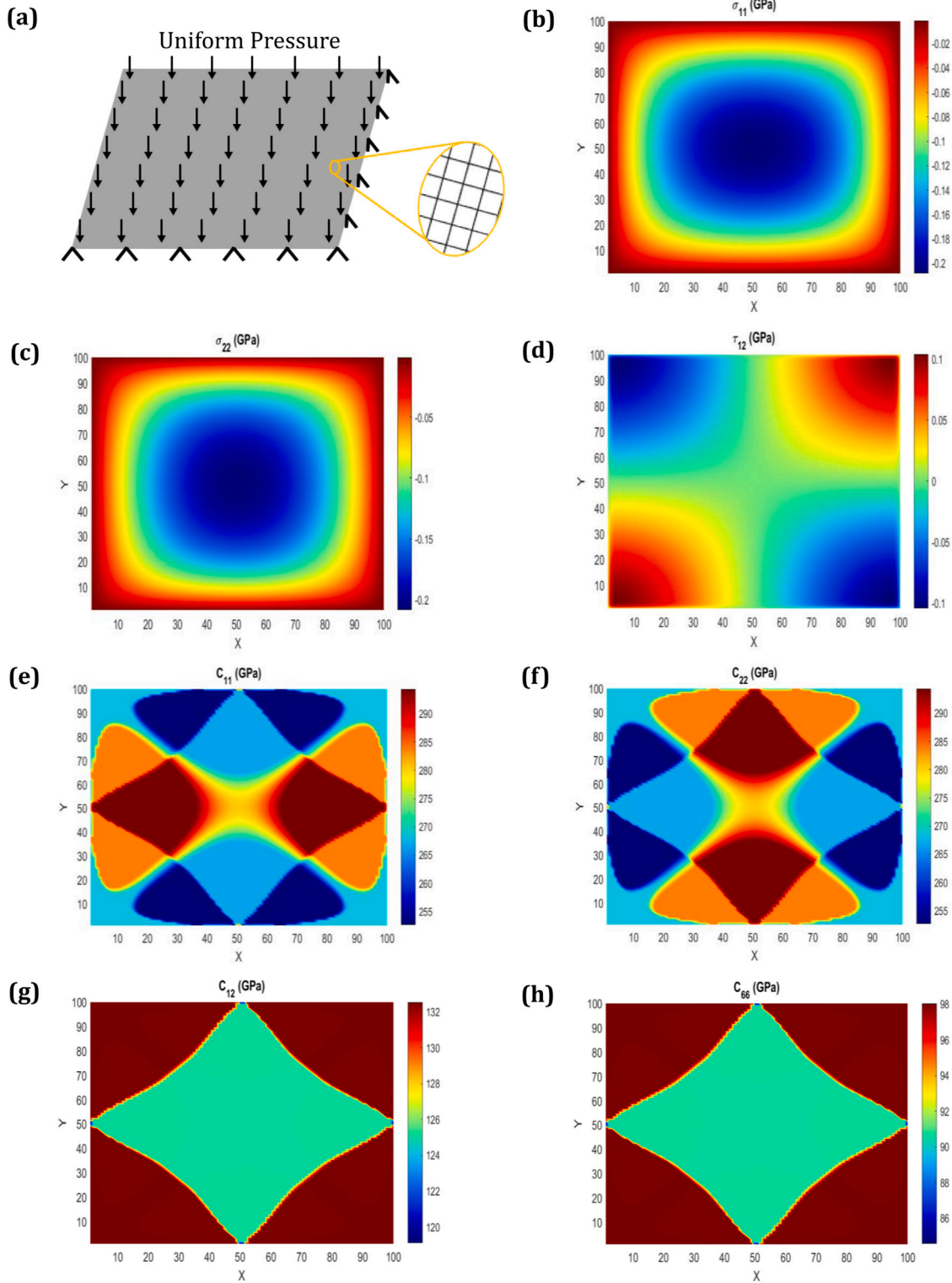


Fig. 10. (a) Schematic diagram of the simply supported square thin plate subjected to a distributed load, along with the mesh discretization; and corresponding stress component distributions over the plate at the initial iteration: (b) σ_{11} , (c) σ_{22} , and (d) τ_{12} . The finalized distributions of optimal properties across the cantilever beam after the 4th iteration: (e) C_{11} , (f) C_{22} , (g) C_{12} , and (h) C_{66} . Dimensions are in 'mm', and stresses and stiffness coefficients are in 'GPa'.

and deformation behavior, with mesh resolution deliberately chosen to precisely resolve optimized property distribution gradients throughout the structures.

Cantilever beam with end load: Beam bending exemplifies plane stress conditions due to small thickness relative to in-plane dimensions,

rendering out-of-plane stresses insignificant. The finite element model represents a two-dimensional cantilever beam (50 mm $\times$ 10 mm $\times$ 1 mm) discretized using triangular Delaunay elements over a 250 $\times$ 50 structured grid. The left edge is fully constrained (fixed support), while a 100 N vertical load is evenly distributed downward along the right edge as shown in Fig. 9(a). The three in-plane stress

components illustrated in Fig. 9(b–d) reveal σ_{11} as the dominant strain energy contributor. As bending moments vary from zero at the free end to maximum at the fixed end, σ_{11} magnitude varies correspondingly across beam height, directly impacting optimized stiffness property distributions as represented in Fig. 9(e–h). The axial stress σ_{11} is zero along the neutral axis and low in its vicinity, causing optimal C_{11} distribution to exhibit a minimum near the beam midline. Conversely, shear stress τ_{12} is higher near the neutral axis, leading to elevated optimal C_{12} and C_{66} values in that region. This interplay produces a tapered cone-shaped pattern in optimized stiffness properties, with cone thickness diminishing progressively with increasing bending moment along beam length, reflecting localized material property adaptation to varying stress environments.

Simply supported thin plate with distributed load: The out-of-plane loading case involves bending analysis of a thin square plate (100 mm $\times$ 100 mm $\times$ 1 mm) under plane stress assumption. A uniformly distributed downward pressure of 1/15 MPa is applied over the entire surface as pictured in Fig. 10(a). Simply supported boundary conditions allow edge rotation while enforcing zero transverse displacement ($w = 0$). The plane stress assumption is justified by the thin plate geometry (1 mm thick relative to 100 mm in-plane), making through-thickness stress negligible. The domain uses a 100 $\times$ 100 structured quadrilateral mesh (10,000 elements, 10,201 nodes). Initial iteration in-plane stress components as shown in Fig. 10(b–d) exhibit symmetry about vertical and horizontal mid-planes due to symmetric loading, boundaries, and material properties. The plate divides conceptually into four identical segments with similar stress patterns. This symmetry governs the stress field and directly influences optimal stiffness distribution as outlined in Fig. 10(e–h). Normal stresses σ_{11} and σ_{22} maximize at the center due to bending and diminish toward edges. In-plane shear stress τ_{12} remains negligible along midlines and center but increases toward corners, reflecting typical thin plate response under symmetric loading. Optimal C_{11} and C_{22} distributions form four symmetric regions of Fig. 10(e–f), directly reflect stress field symmetry. Fig. 10(g–h) represents that the shear-related properties C_{12} and C_{66} are lower centrally and higher near corners, corresponding to τ_{12} distribution.

3.2.2. Mesostructure-graded structures

This study involves a mesostructure-graded simply supported beam subjected to a mid-span in-plane load. Since the proposed optimization method for microstructure-graded and mesostructure-graded structures is identical, and multiple example cases have already been presented in earlier sections for microstructure-graded structures, only one representative example is considered here. This example is further validated through both numerical simulations and experimental procedures.

The schematic diagram of the considered beam case is illustrated in Fig. 3(b), which also depicts the placement of the honeycomb mesostructure within the plane beam section. The finite element model represents only the left half of a simply supported beam, using symmetry about the midpoint to reduce computational cost. The modeled section is 50 mm long, 10 mm high, and 1 mm thick. A uniformly distributed downward load of 0.10 N is applied at the right end ($x = 50$ mm). Boundary conditions are: zero vertical displacement ($u_y = 0$) at the left end ($x = 0$ mm) and zero horizontal displacement ($u_x = 0$) at the right end ($x = 50$ mm), enforcing mirror symmetry. This setup corresponds to a full 100 mm beam under a 0.20 N central point load. The mesh generation scheme remains consistent with the previous cantilever beam case. However, the present mesostructure-graded beam design accounts for both UD and NUD distribution scenarios. Fig. 11(a–d) illustrates the optimal property distribution across the left half of the beam, considering a density of 45%. The observed trends are closely similar to those of the microstructure-graded beam case, where the core region of the beam tends to achieve higher shear and transverse tensile moduli (C_{12} , C_{66} , and C_{22}), while the outer region aims to attain higher longitudinal tensile stiffness (C_{11}). However, these higher and

lower values do not necessarily correspond to the absolute maximum or minimum of the respective stiffness components. The optimal values depend on the specific shape of the property space.

On the other hand, Fig. 11(e–h) illustrates the optimal property distribution across the left half of the beam, considering a varying density, ranging from 30% to 60%, with ensuring an average of 45%. This optimal property distribution exhibits a markedly different trend, as several distinct and diverse regions can be observed. Furthermore, the optimal C_{12} and C_{66} distributions display opposite behaviors, remaining completely distinguishable from the earlier cases.

The smooth spatial gradients in material properties across the design domain, which are continuous and closely aligned with the stress distribution along the structure, adapt to the load paths. Traditional FGMs utilize predefined mathematical functions to model material property variations. In contrast, this approach derives the material distribution through an iterative optimization process that incorporates structural deformation and local energy minimization. Unlike traditional methods, which assume a uniform gradient, proposed method produces gradients based on the structural load path and performance requirements. This inverse-design approach optimizes strain energy directly, eliminating the need for predefined functions.

3.3. Transitions

The transition of microstructures and mesostructures is crucial for ensuring the manufacturability and applicability of FGMs. However, both types of optimal structures are primarily illustrated through essential graphical representations, while mesostructure-graded examples are further extended and supported through experimental testing.

3.3.1. Transition of microstructure

This subsection focuses on tracking optimal property point positions within the three-dimensional property closures of C_{11} , C_{22} , and C_{12} . This tracking process is crucial for capturing the transition of microstructure and mechanical properties for designing the FGM structures and understanding the influence of constraints that govern the resulting solutions. Among all considered scenarios, two representative loading cases are explored in detail: one involving in-plane loading, represented by a cantilever beam subjected to an end load, and another involving out-of-plane loading, represented by a simply supported thin plate under distributed load. These cases are illustrated in Fig. 12. To effectively visualize the distribution of optimal points within the property space, black discontinuous lines are used to demarcate the symmetric regions in both structural configurations. For the cantilever beam, symmetry in optimal property distribution is defined by a single neutral axis, which splits up the structure into two symmetric zones. Although the strain and stress distributions are asymmetric, with tension in the upper region and compression in the lower, their equal and opposite magnitudes lead to a symmetric distribution of properties. Conversely, the thin plate under bending exhibits four symmetric regions, as previously discussed in an earlier subsection.

The example structures contain a large number of elements in their finite element analyses. As a result, the corresponding optimal property points, when plotted along with three-dimensional stiffness property space, appear cluttered and indistinguishable, making it difficult to interpret their relative distribution. To address this issue, the property space is partitioned into smaller sub-regions, referred to as bins. To enhance the clarity of visualization, a metric termed optimal point density is introduced, which represents the number of optimal points contained within each bin. In Fig. 12, the optimal point density is illustrated through varying radii of black spheres, where lower density corresponds to smaller radii and higher density corresponds to larger radii, effectively conveying the relative distribution of optimal points across the space. Among all four stiffness coefficients, the optimal distribution of C_{11} remains the most diverse across all examples, as observed in the previous subsection. Consequently, this component has

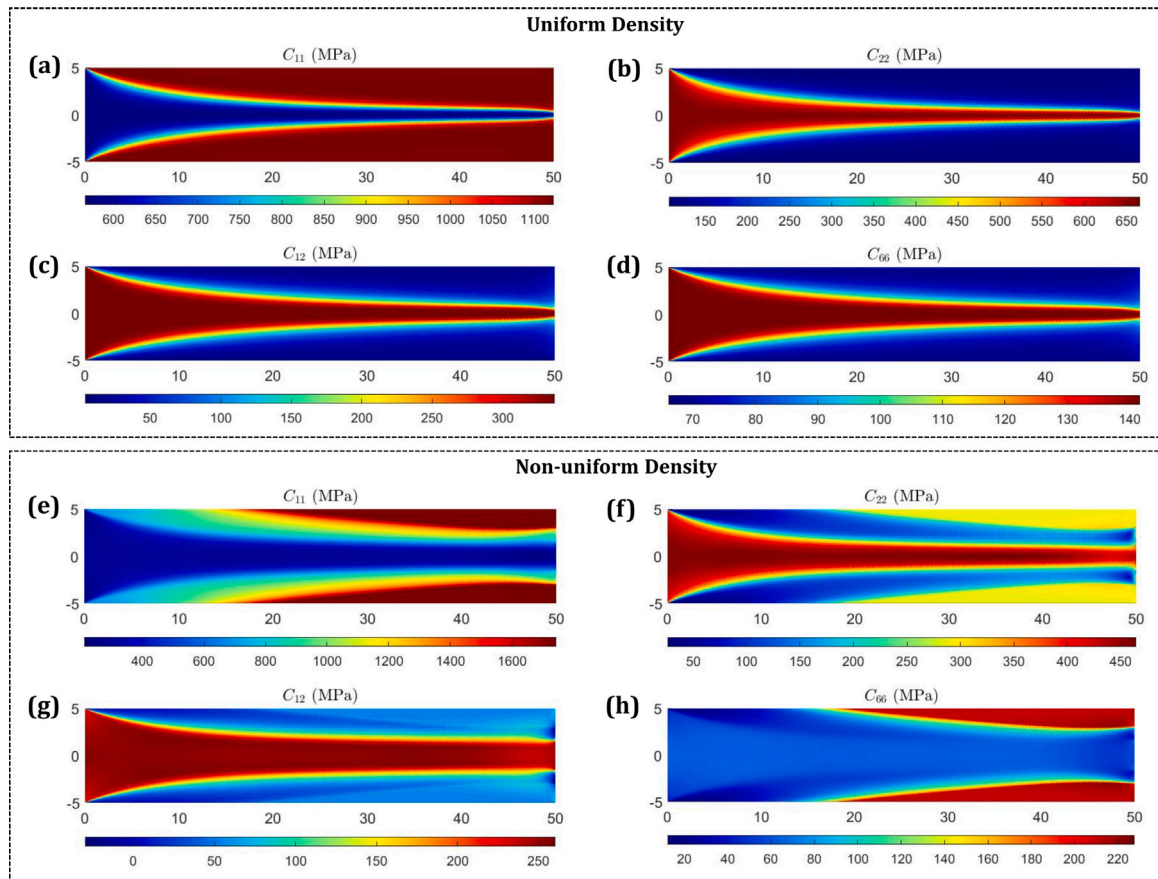


Fig. 11. Finalized distributions of optimal properties across left half of the simply supported beam under the conditions shown in Fig. 3(b) after the 4th iteration: for UD (a) C_{11} , (b) C_{22} , (c) C_{12} , and (d) C_{66} ; for NUD (e) C_{11} , (f) C_{22} , (g) C_{12} , and (h) C_{66} . Dimensions are in 'mm', and stiffness coefficients are in 'MPa'.

been selected for detailed analysis, and its entire distribution is divided into five major regions for both example structures. These regions contain a greater number of elements, resulting in a higher density of corresponding points in the property space, which in turn form larger-radius spheres. Although the specific locations of these five dense regions differ between the two examples, several consistent patterns can be observed:

- (i) Almost all optimal points are located on the upper side of the property space.
- (ii) Each optimal point lies within at least one constraint surface, and most reside within two or more.
- (iii) All dense points are positioned at the vertices of the property space and are interconnected by a large number of lower-density points.
- (iv) The optimal points are nearly continuously connected, mirroring the connectivity of their corresponding elements in the physical structure.
- (v) Overall, the five dense regions are circularly connected which form a loop-like pattern: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 1$.

These five key observations support the applicability of the FGM structure concept in achieving optimized stiffness distribution and structural performance.

A comprehensive analysis reveals that six optimal property points within the closure domain play a dominant role in shaping the optimal microstructures identified in both cases under consideration. Subsequently, the corresponding dominant microstructural textures are inversely determined and represented within Rodrigues space and essential pole figures, as illustrated in Fig. 13. To reconstruct the

ODFs corresponding to these six dominant property points (labeled 1 through 6), a constrained nonlinear optimization scheme is employed using a composite objective function. This optimization minimizes the deviation between the target and homogenized stiffness coefficients $\{C_{11}, C_{22}, C_{12}\}$, computed as a linear combination of constituent stiffness tensors weighted equally for these three components. The volume fraction vector is enforced as an equality constraint to preserve the probabilistic consistency of the ODF, as defined earlier in Eq. (2). The optimization bounds are set between zero and a large upper limit to allow for single crystal orientations, with the initial guess set at the lower bound. A high-precision gradient-based optimization is performed using the interior-point algorithm, incorporating stringent tolerances for step size, optimality, and constraint satisfaction to ensure accurate convergence. The final results are visualized through the Rodrigues space of the cubic crystal system and corresponding pole figure plots. To aid interpretation, a schematic representation of the Rodrigues space with dependent and independent orientation nodal points is provided in Fig. 13(a). The six dominant optimal property points, labeled sequentially from 1 to 6 within the property closure, are shown in Fig. 13(b), while their associated texture representations are depicted in Fig. 13 (rows 1 to 6). Among all the considered structures, each contains thousands of elements, making it challenging to effectively visualize the corresponding microstructures within the scope of this discussion. To manage this complexity, only the most significant dominant optimal points are selected for detailed analysis. As a result, the transitional microstructures between these dominant points are not presented. Overall, the visualizations of these dominant optimal microstructure textures reveal that they are mostly sharp, as they are derived from the vertex points of the property closure. For instance, the microstructure corresponding to optimal property point

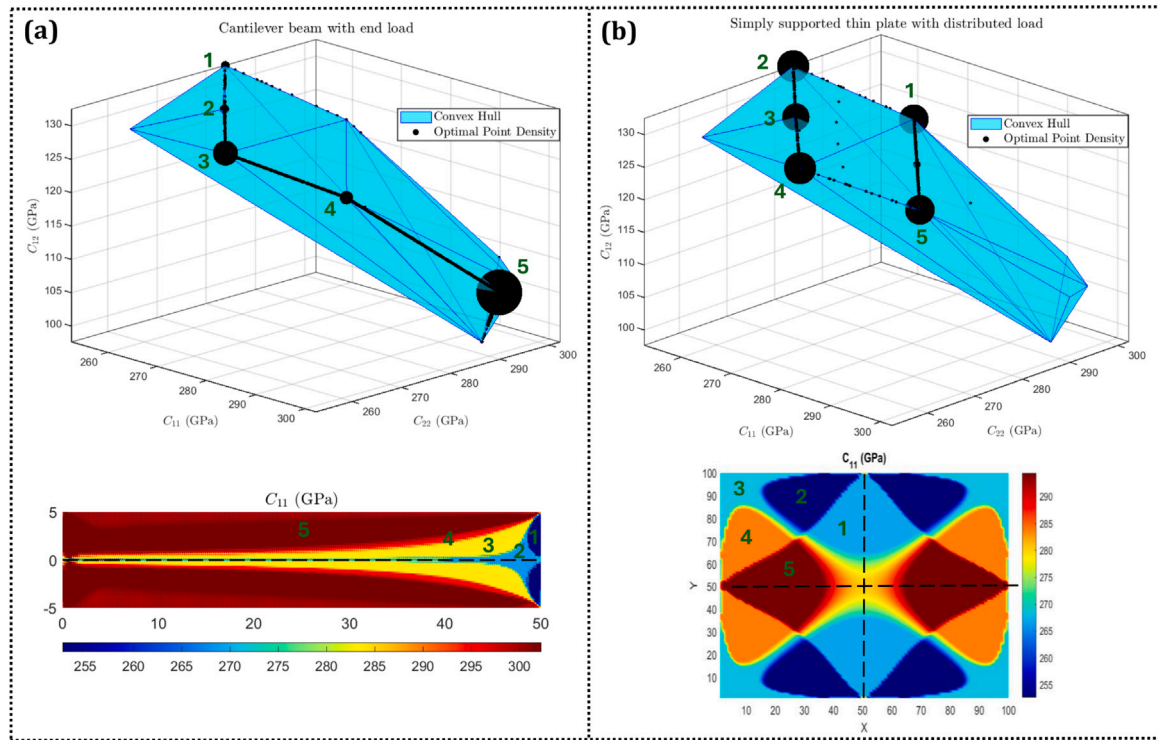


Fig. 12. Optimal property point locations in property spaces for two structural examples: (a) a cantilever beam with an end load and (b) a simply supported thin plate with a distributed load. The black spheres indicate the optimal points, and their sizes represent the relative density. Larger spheres correspond to regions with higher density.

number 3 exhibits only three prominent orientations, located at node numbers 1, 33, and 34 in Rodrigues orientation space coordinates of $(0, 0, 0)$, $(0, \sqrt{2} - 1, 0)$, and $(\sqrt{2} - 1, 0, 0)$ respectively. Among these, nodes 33 and 34 which lie on the boundary of the Rodrigues space, have a dominant influence on the microstructure, as illustrated in Fig. 13 (3). This indicates that the grains within the microstructure have the highest probability of being oriented at angles of $\varphi_x = 0^\circ$, $\varphi_y = 2 \tan^{-1}(\sqrt{2} - 1)$, $\varphi_z = 0^\circ$, or $\varphi_x = 2 \tan^{-1}(\sqrt{2} - 1)$, $\varphi_y = 0^\circ$, $\varphi_z = 0^\circ$. These angles result from the use of the axis-angle parameterization in Rodrigues orientation space, where the rotation vector is defined as $\mathbf{r} = \mathbf{n} \tan(\varphi/2)$, with $\mathbf{n}$ representing the rotation axis and φ the angle of rotation. However, the transitions between two successive optimal microstructures are generally continuous in nature, indicating gradual changes in the ODF values at intermediate points.

This key observation highlights that the microstructures change continuously, much like physical elements are connected to one another. The main difference lies in the varying gradients of these changes, which are largely influenced by the stress distribution or load path within a given structure. These findings support the applicability of FGM structures in achieving optimized stiffness distributions and in reducing deformation under applied loading conditions.

3.3.2. Transition of mesostructure

The transition of mesostructure within the optimal beam design is illustrated through the variation of geometric properties of the honeycomb unit cell. Fig. 14(a–c) present the distributions of geometric parameters such as l/h , θ , and t/h under the assumption of UD across all unit cells. Subsequently, Fig. 14(d–g) show the corresponding distributions under NUD conditions, which additionally include the density (ρ) distribution. In both cases, the optimal ratios of inclined and vertical rib lengths converge to unity across all elements. Meanwhile, the other two geometric variables: the inclination angle (θ) and the rib thickness-to-vertical rib length ratio exhibit diverse distributions. The

most notable observation arises from the inclination angle distribution, which appears unaffected by geometric constraints. Although the constraints limit θ between -25° and 60° , the optimal distribution predominantly clusters around either 0° or 30° . This indicates that the optimal unit cell tends to form either a nearly rectangular or a nearly perfect hexagonal honeycomb, aside from the transition regions between them. In the case of UD, the rib thickness-to-vertical rib length ratio acts as a dependent variable whose variation follows the trend of relative density. However, under NUD, this ratio becomes an independent variable, as the added density flexibility introduces an additional degree of freedom. The resulting density distribution reveals a lighter region near the neutral axis and denser regions farther from the neutral axis, particularly near the load application points. Finally, Fig. 14(h) illustrates the computer-aided design (CAD) models of both the reference and optimized beams.

3.4. Performance improvement

This subsection presents a comparative analysis of the physical performance of the studied structures before and after stiffness optimization, as governed by microstructural and mesostructural property constraints. The primary objective of the optimization process is the minimization of strain energy, which fundamentally enhances the overall stiffness of the structures. To evaluate the impact of this optimization, vertical deflection has been selected as the key performance indicator, as it provides the most prominent insight into structural behavior under the applied loads.

3.4.1. Microstructure-graded structure

Fig. 15 illustrates the vertical deflection profiles for in-plane and out-of-plane loading cases. Given that the external loads are applied in a direction that induces vertical displacement, analyzing vertical deflection offers a direct and meaningful assessment of structural response and improvement.

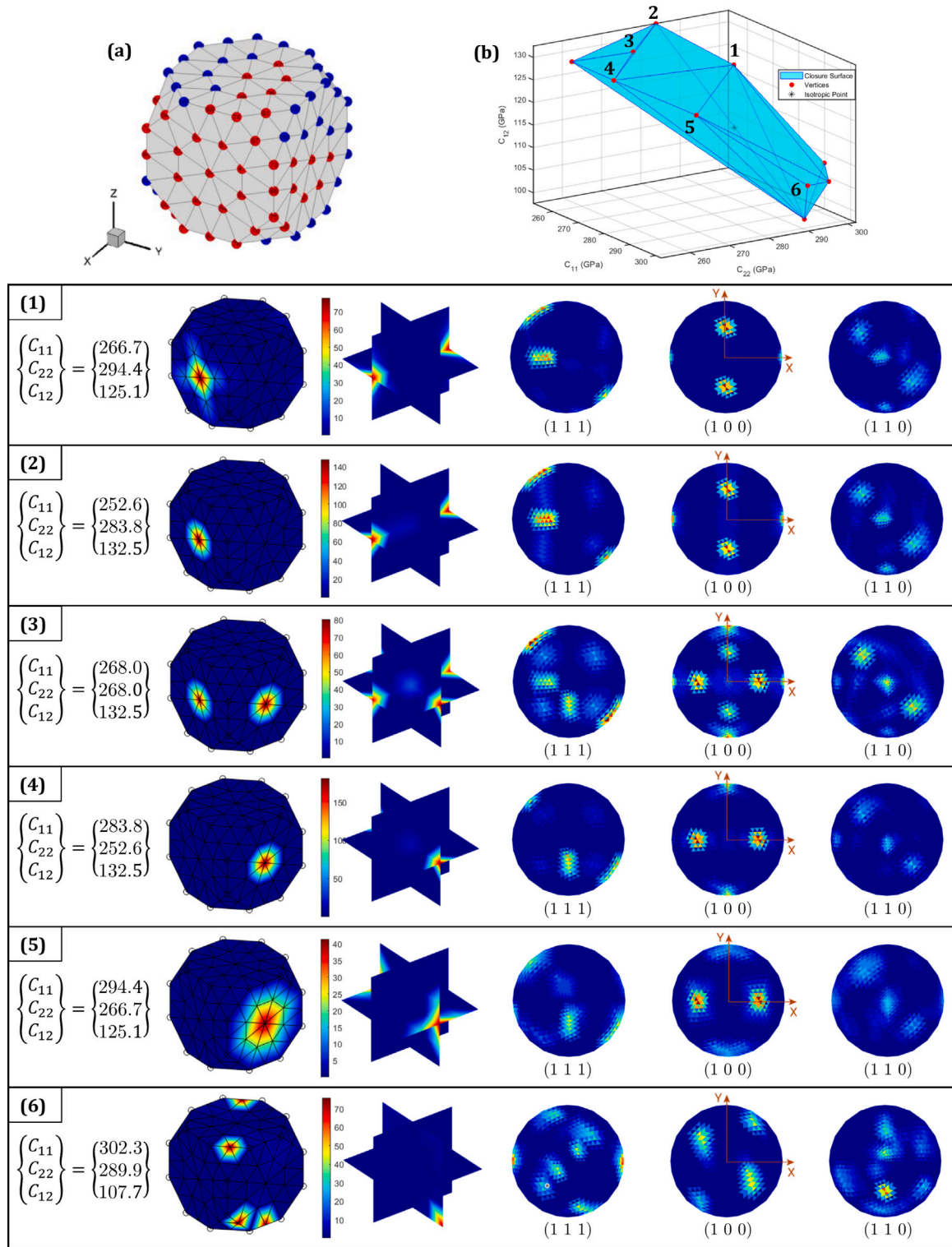


Fig. 13. (a) Discretized Rodrigues orientation space of the FCC system, where nodal points represent ODFs. Red points indicate independent orientations, while blue points represent dependent orientations. (b) Dominant optimal property points within the closure, along with their corresponding microstructures labeled as 1 to 6. (1–6) Representation of the respective microstructures using Rodrigues orientation space and the corresponding three pole figures. Stiffness coefficients are in ‘GPa’.

For the cantilever beam cases, the external load is applied at the free end of the structure, resulting in peak deflection at the point of load application, as shown in Fig. 15(a–c). In contrast, the thin square plate is subjected to a uniformly distributed load with simply supported boundaries along all edges, leading to maximum deflection

at the center of the plate, as illustrated in Fig. 15(d–f). In both in-plane (beam) and out-of-plane (plate) loading scenarios, the optimized distributions of material properties produce observable improvements in structural stiffness, reflected by considerable reductions in maximum deflection. Specifically, the maximum deflection decreases by

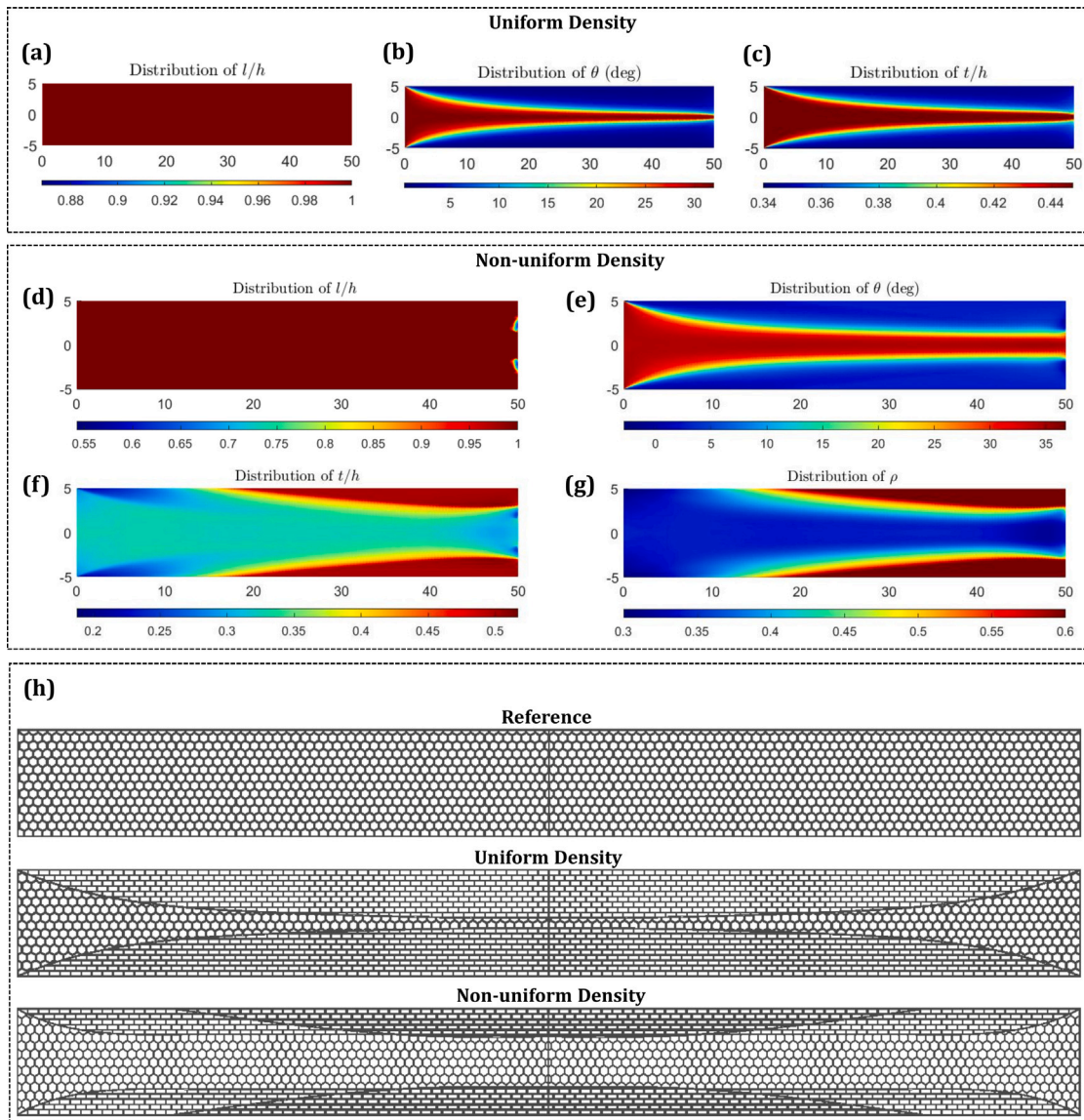


Fig. 14. Optimal geometric parameter distributions of the honeycomb unit cell mesostructure for the simply supported beam: (a–c) correspond to the case with a constant density of 45%; (d–g) correspond to the case with a varying density between 30% and 60%, ensuring an average density of 45%; (h) shows the streamlined CAD models used for numerical (Abaqus) and experimental validations.

approximately 10.5% for the cantilever beam and 7.9% for the thin plate under distributed loading. These comparative results are obtained by comparing the optimal performances against a reference case with a randomly oriented microstructure uniformly distributed throughout the structures. The corresponding property values of this reference microstructure are also taken as the initial design properties for all the example structures considered. These reductions confirm the effectiveness of the optimization process in enhancing load-bearing performance without altering the fundamental deformation behavior. Although the magnitudes of deflection decrease, the overall deflection shapes remain consistent across cases, indicating that the optimization primarily scales down deformation intensity proportionally throughout the finite element domain. It is also evident that most of the improvement occurs after the first optimization iteration, while subsequent refinements yield progressively smaller gains, as seen in Fig. 15(c) and (f). This behavior arises because the strain energy minimization function depends on stress components from the previous iteration, which slightly adjust as material stiffness is updated. Consequently,

an iterative finite element procedure ensures convergence toward an optimal configuration, with diminishing changes in later iterations.

3.4.2. Mesostructure-graded structure

Fig. 16 illustrates the performance improvement of the mesostructure-graded beam in terms of vertical deflection. The homogenization approach determines the optimal mesostructure distribution within the beam, which is further validated numerically (using Abaqus) and experimentally. Specifically, the homogenization approach encompasses all optimization iterations, whereas the numerical and experimental investigations focus only on the initial and final optimized beam performances. The initial mesostructure within the beam is assumed to have a uniform 45% density with a perfect hexagonal honeycomb configuration in both uniform density (UD) and non-uniform density (NUD) cases. Fig. 16(a) depicts the maximum deflection response across all considered scenarios, ultimately revealing the trend associated with microstructure grading. The UD case demonstrates faster convergence compared to the NUD case. The optimal beam structure

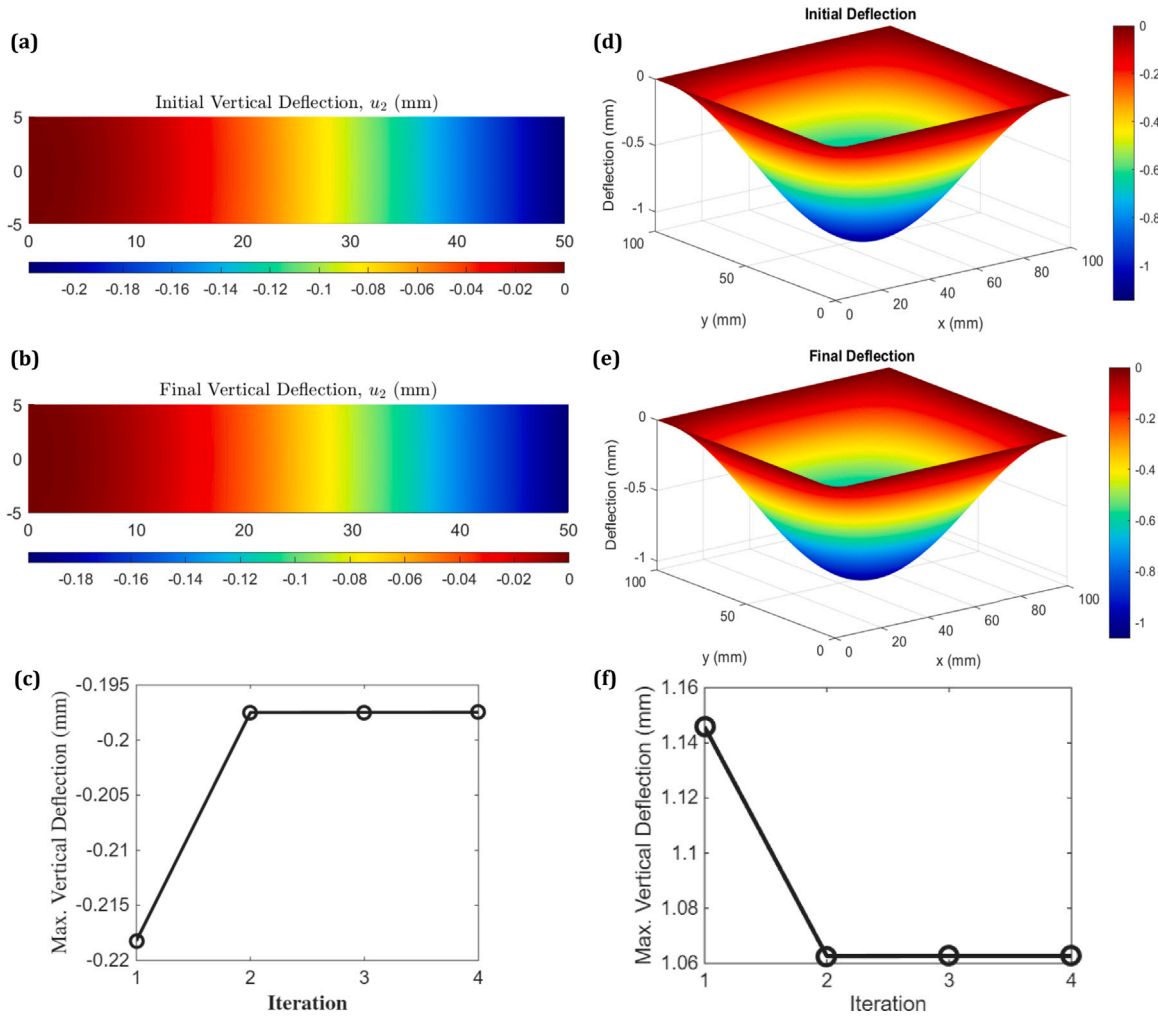


Fig. 15. Comparative vertical deflection behavior of in-plane and out-of-plane loaded structures. (a–c) Cantilever beam under end load; (d–f) Simply supported thin plate under distributed load. (a, d) reference model with uniform isotropic properties; (b, e) optimized design with FGM orthotropic properties; (c, f) deflection response versus optimization iterations.

CAD model is reconstructed based on the optimal geometry distribution obtained from the homogenization process. The Abaqus simulations are performed using identical geometrical dimensions to those used in the homogenization approach. The complex geometric shapes of the optimal structures require an appropriate element choice to accurately capture the mechanical response. Therefore, we employ the C3D20 20-node quadratic hexahedral element, which is widely regarded as a robust and reliable choice for three-dimensional beam-like structures. Since the Abaqus simulations are used solely for validation purposes and are not involved in the optimization process, high accuracy and convergence are maintained without compromising the element type or mesh size. Fig. 16(b) and (c) depict the local deflection profiles from the homogenization and numerical schemes, respectively, showing how effectively the current homogenization-based FEM captures the behavior of the real mesostructured beam.

In the experimental three-point bending setup, the beam specimens are scaled up by 2.5 times in the in-plane dimensions and 10 times in the out-of-plane dimension relative to the original CAD files. This scaling demonstrates their printability using a Bambu X1C 3D printer and ensures an acceptable unit cell size and smooth transitions. To compare the experimental results, the equivalent flexural modulus is calculated from the applied force–deflection curve (within the 3 mm linear region), as illustrated in Fig. 16(d). This equivalent flexural

modulus is subsequently used to determine the beam response for the original dimensions. Fig. 16(e) presents a representative image of the experimental setup, further justifying the scaling approach to align with standard bending tests. While considering the original homogenization approach, the NUD case exhibits an approximately 62% reduction in maximum deflection, whereas the UD case shows an approximately 54% reduction in maximum deflection, compared with the initial or reference design (a uniform hexagonal honeycomb unit cell distributed throughout the entire beam). Despite minor deviations among the three methods, the overall improvement trends remain consistent, confirming that the NUD optimal structure outperforms the UD beam configuration. The use of similar models in both the numerical simulations and experimental setup validates the homogenization approach.

4. 3D extension of this study

The major focus of this study is the design of a 2D property space that captures the closure of associated stiffness components. The 2D structure itself is simpler to illustrate and analyze; however, the methodology can be naturally extended to 3D structures. The IN-718 alloy, known for its wide application in pressurized fluid transfer systems, often requires the design of dimensionally stable elbow components. Therefore, this section analyzes the extension of the current

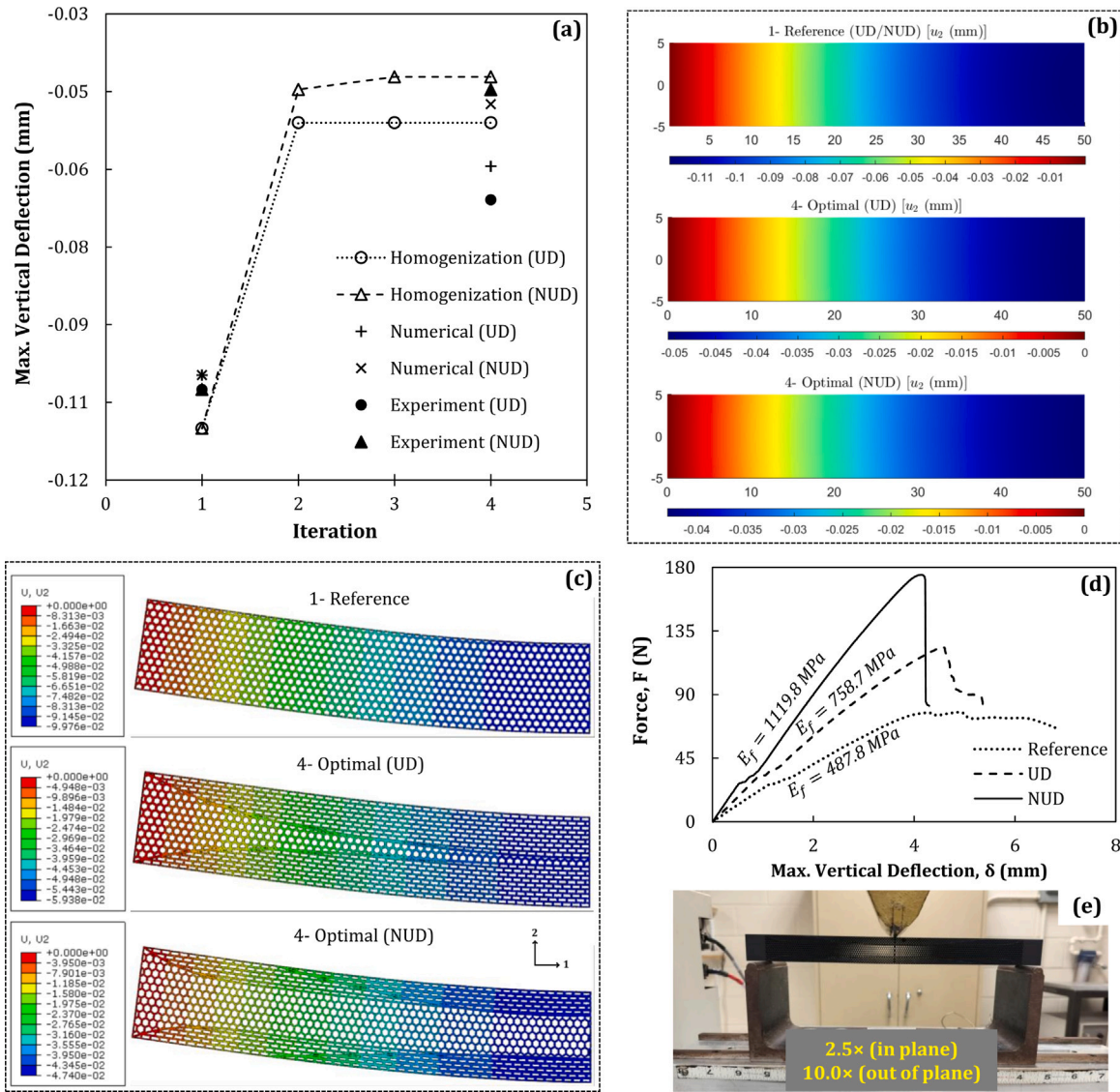


Fig. 16. (a) Comparative analysis of maximum vertical deflection of the beam obtained from different investigations; (b) initial (reference) and optimal deflection of the left half of the beam based on the homogenization method; (c) reference and optimal deflection of the left half of the beam obtained from the numerical (Abaqus) method; (d) reference and optimal maximum deflection of the beam obtained from the experimental method; (e) three-point bending test setup with the scaling factors of the specimens relative to the original dimensions.

2D method to 3D structures by investigating the microstructure-graded elbow element design. The proposed approach can readily be applied to form a microstructure-informed property space, which then guides the design of microstructure-graded 3D structures. By identifying the vertex orientations and their corresponding property points, both inequality and equality constraints for all orthotropic stiffness components are established (listed in ‘Supplementary Materials’). Subsequently, a generalized optimization framework is employed to determine the element-specific optimal property distribution. The objective function remains the same as that defined in Eq. (15), but it is expanded for 3D elements to include all stiffness and stress components, as formulated in Eq. (19) (see the Eq. (19) in Box I).

Following this, inverse microstructure calculations are performed to obtain element-specific microstructures. As an illustrative example, a simplified elbow geometry with an inner radius of 15 mm, outer radius of 20 mm, and curvature radius of 45 mm is analyzed under an internal pressure of 100 MPa, as shown in Fig. 17(a). The corresponding von Mises stress distribution, computed using Abaqus, is presented in Fig. 17(b). Two representative nodes are selected: the most probable

failure node (point A) and the least probable failure node (point B), for which the optimal microstructures are determined. The six stress components of these nodes are shown in Fig. 17(c) and are used to derive the optimal microstructures, visualized in Fig. 17(d) through the Rodrigues orientation space and the three fundamental pole figures. The shown microstructures correspond to iteration 2 of the optimization process; although further iterations may induce minor adjustments, significant convergence is already achieved by this stage. Moreover, a similar approach can be extended to design mesostructure-graded 3D components, incorporating three-dimensional unit cells such as Triply Periodic Minimal Surface (TPMS) architectures and their associated property spaces.

5. Conclusion

This study introduces an efficient method for evaluating property closures derived from FCC microstructural information, presenting a novel texture-driven approach to effectively assess property spaces. The

$$f_{3D}(C) = \frac{(C_{22}C_{33} - C_{23}^2)\sigma_{11}^2 + (C_{11}C_{33} - C_{13}^2)\sigma_{22}^2 + (C_{11}C_{22} - C_{12}^2)\sigma_{33}^2}{\Delta} + \frac{2(C_{13}C_{23} - C_{12}C_{33})\sigma_{11}\sigma_{22} + 2(C_{12}C_{23} - C_{13}C_{22})\sigma_{11}\sigma_{33} + 2(C_{12}C_{13} - C_{11}C_{23})\sigma_{22}\sigma_{33}}{\Delta} + \frac{\tau_{23}^2}{C_{44}} + \frac{\tau_{13}^2}{C_{55}} + \frac{\tau_{12}^2}{C_{66}}, \quad \text{where } \Delta = \det \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{12} & C_{22} & C_{23} \\ C_{13} & C_{23} & C_{33} \end{bmatrix}. \quad (19)$$

Box I.

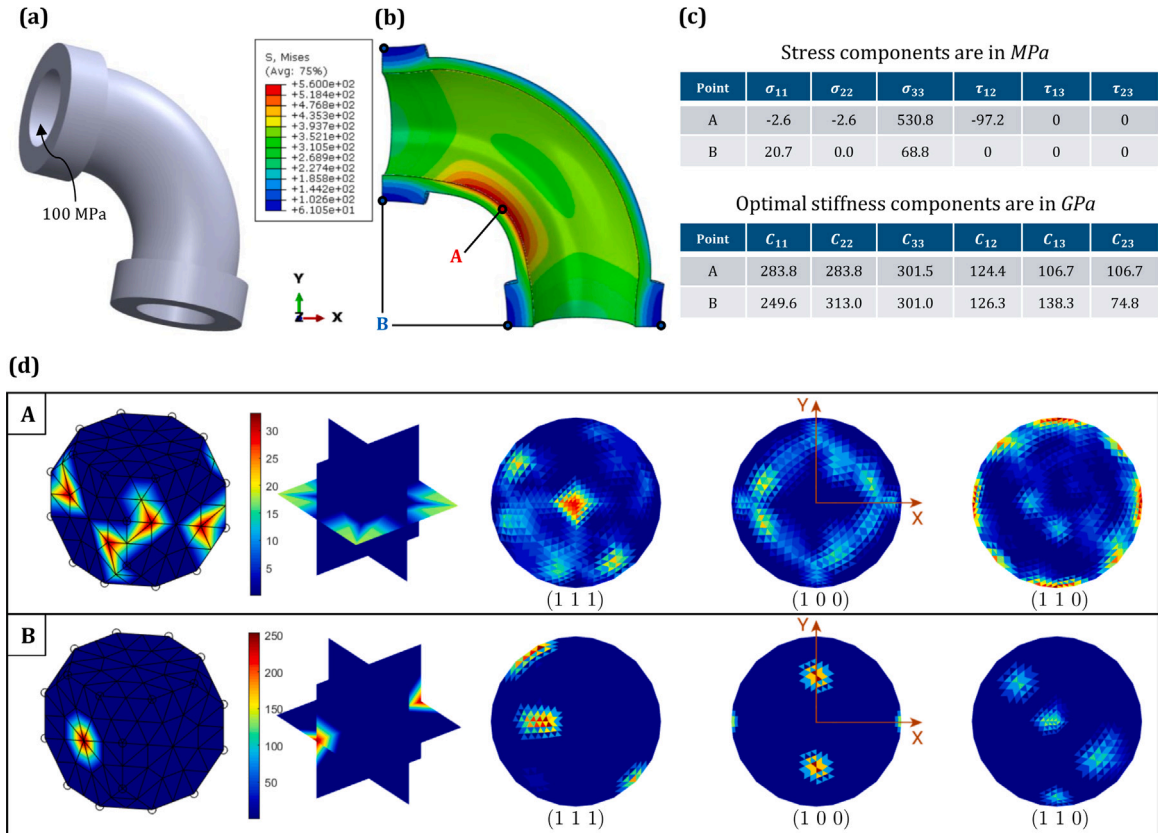


Fig. 17. (a) Example of the pressurized elbow considered in this study; (b) von Mises stress distribution on half of the elbow cross-section; (c) stress components at two representative points and their corresponding optimal stiffness properties; (d) representation of the respective microstructures in the Rodrigues orientation space along with their associated three pole figures.

mathematical modeling of Malek and Gibson is incorporated to construct mesostructure-informed property closures for honeycombs. The presented property-space generation approach is significantly less computationally expensive than conventional data-driven methods while remaining straightforward to apply in structural design. Furthermore, the study reveals how feature-level DOF at different material scales influence the resulting property-space geometry, demonstrating robust connectivity between material variables and attainable properties across both micro and meso scales.

The immediate applicability of both property closures to the design of FGMs has also been demonstrated in a generalized manner. This straightforward strain-energy-based iterative optimization framework is compatible with all finite element types, as illustrated through multiple examples involving 2D plane-stress conditions under in-plane and out-of-plane loading, as well as 3D solid element structures. This enables enhanced structural performance without requiring macroscopic geometric modifications. Notably, the microstructure-graded

beam exhibits approximately 11% reduction in deflection, while the mesostructure-graded non-uniform-density beam achieves around 62% reduction relative to corresponding baseline structures.

Moreover, a detailed analysis of stiffness-property transitions, corresponding microstructural textures within the feasible property space, and mesostructural geometric and density distributions underscores the continuity and practicality of the optimized designs. These results highlight the strategic alignment of optimized material distributions with stress paths and connectivity patterns that reflect physical element interactions. The effectiveness of this method is validated through numerical simulations and experimental testing of mesostructure-graded specimens, with results showing consistent agreement with the proposed homogenization approach.

However, the limitations of this study stem from the use of stress fields solely in the calculation of strain energy, which often leads to stress concentrations that can reduce the load-bearing capacity. Future work should focus on incorporating composite objectives that account

for maximum stresses (such as von Mises or maximum shear, depending on the material type), alongside compliance. Additionally, multi-physics objectives, including thermomechanical coupling and nonlinear behavior beyond the elastic limit of the base material, should be explored, particularly given the versatile applications of metamaterials. Process-aware manufacturing constraints, both at the micro and meso scales, should also be considered to advance material-informed structural design. Furthermore, the combined analysis of micro- and mesostructural variations can potentially expand the property space, offering new avenues for study and extending the approach to macro-scale topology optimization.

CRedit authorship contribution statement

Md Maruf Billah: Writing – review & editing, Writing – original draft, Validation, Methodology, Data curation, Conceptualization. **Waris Khan:** Writing – review & editing, Conceptualization. **Pinar Acar:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Analytical equations for honeycombs

Malek and Gibson [61] method establishes the effective elastic moduli and Poisson’s ratios in terms of the geometric parameters of the honeycomb unit cell as shown in Fig. 3(b); specifically, the wall thickness (t), effective bending lengths (l_b, h_b), cell angle (θ), and wall aspect ratio (h/l) along with the elastic constants of the isotropic base material (E_s, ν_s). The effective lengths are defined as, $l_b = l - \frac{t}{2 \cos \theta}$, $h_b = h - \frac{t(1 - \sin \theta)}{\cos \theta}$. Eqs. (A.1)–(A.5) define the directional Young’s moduli (E_1, E_2), Poisson’s ratios (ν_{12}, ν_{21}), and the in-plane shear modulus (G_{12}), whereas Eq. (A.6) introduces the auxiliary parameter K , which governs shear compliance through coupled bending and axial deformation of the cell walls. Collectively, these relationships capture the geometric nonlinearity and mechanical coupling induced by wall inclination and nodal interactions, thereby enabling accurate prediction of the in-plane orthotropic behavior of periodic hexagonal honeycombs across a wide range of relative densities. The relative density (ρ) of the periodic hexagonal honeycomb is expressed as a function of the geometric parameters of the unit cell as shown in Eq. (A.7). This formulation captures the reduction in relative density resulting from nodal geometry and wall inclination, thereby providing a more realistic representation of the material’s effective stiffness at finite wall thickness ratios. Later, the stiffness matrix (C) is evaluated under the plane stress condition through Eq. (A.8), facilitating the determination of the effective in-plane elastic response for subsequent property closure development.

$$E_1 = E_s \left(\frac{t}{l_b} \right)^3 \frac{\cos \theta}{(h/l + \sin \theta) \sin^2 \theta} \left[\frac{1}{1 + (2.4 + 1.5\nu_s + \cot^2 \theta) \left(\frac{t}{l_b} \right)^2} \right], \quad (\text{A.1})$$

$$E_2 = E_s \left(\frac{t}{l_b} \right)^3 \frac{(h/l + \sin \theta)}{\cos^3 \theta} \times \left[\frac{1}{1 + (2.4 + 1.5\nu_s + \tan^2 \theta + \frac{2(h_b/l_b)}{\cos^2 \theta}) \left(\frac{t}{l_b} \right)^2} \right], \quad (\text{A.2})$$

$$\nu_{12} = \frac{\cos^2 \theta}{(h/l + \sin \theta) \sin \theta} \left[\frac{1 + (1.4 + 1.5\nu_s) \left(\frac{t}{l_b} \right)^2}{1 + (2.4 + 1.5\nu_s + \cot^2 \theta) \left(\frac{t}{l_b} \right)^2} \right], \quad (\text{A.3})$$

$$\nu_{21} = \frac{\sin \theta (h/l + \sin \theta)}{\cos^2 \theta} \left[\frac{1 + (1.4 + 1.5\nu_s) \left(\frac{t}{l_b} \right)^2}{1 + (2.4 + 1.5\nu_s + \tan^2 \theta + \frac{2(h_b/l_b)}{\cos^2 \theta}) \left(\frac{t}{l_b} \right)^2} \right], \quad (\text{A.4})$$

$$G_{12} = E_s \left(\frac{t}{l_b} \right)^3 \frac{(h/l + \sin \theta)}{\left(\frac{h_b}{l_b} \right)^2 \cos \theta} \left(\frac{1}{K} \right), \quad (\text{A.5})$$

$$K = \left[1 + 2 \left(\frac{h_b}{l_b} \right) + \left(\frac{t}{l_b} \right)^2 \left(\frac{2.4 + 1.5\nu_s}{h_b/l_b} \right) \left(2 + \frac{h}{l} + \sin \theta \right) + \frac{(h/l + \sin \theta)}{\left(\frac{h_b}{l_b} \right)^2} \left[(h/l + \sin \theta) \tan^2 \theta + \sin \theta \right] \right], \quad (\text{A.6})$$

$$\rho = 1 - \frac{l_b(h_b + l_b \sin \theta)}{l(h + l \sin \theta)}, \quad (\text{A.7})$$

$$C = \begin{bmatrix} C_{11} & C_{12} & 0 \\ C_{12} & C_{22} & 0 \\ 0 & 0 & C_{66} \end{bmatrix} = \begin{bmatrix} \frac{E_1}{1 - \nu_{12}\nu_{21}} & \frac{\nu_{12}E_2}{1 - \nu_{12}\nu_{21}} & 0 \\ \frac{\nu_{21}E_1}{1 - \nu_{12}\nu_{21}} & \frac{E_2}{1 - \nu_{12}\nu_{21}} & 0 \\ 0 & 0 & G_{12} \end{bmatrix}. \quad (\text{A.8})$$

Supplementary materials

Additional supporting information and data for this study are available online. These materials include: (i) the complete set of stiffness property constraints used in the optimization, including both inequality constraints and the equality constraints for the IN-718 property closure; (ii) the property closure of the double-arrowhead mesostructure; (iii) detailed discussions of the microstructure-graded simply supported beam and the microstructure-graded L-shaped frame; and (iv) extended numerical results for two additional case studies: a tapered cantilever beam subjected to an end load (in-plane loading scenario) and a clamped thin plate under a uniformly distributed load (out-of-plane loading scenario). All supplementary content can be accessed via the following GitHub repository:

<https://github.com/MARUF087/Generalized-Approach-to-Designin-g-Functionally-Graded-Materials.git>

Data availability

A portion of the data has been made available on GitHub, and the remaining data can be provided upon request.

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